



2026 Level 2 - Seminars

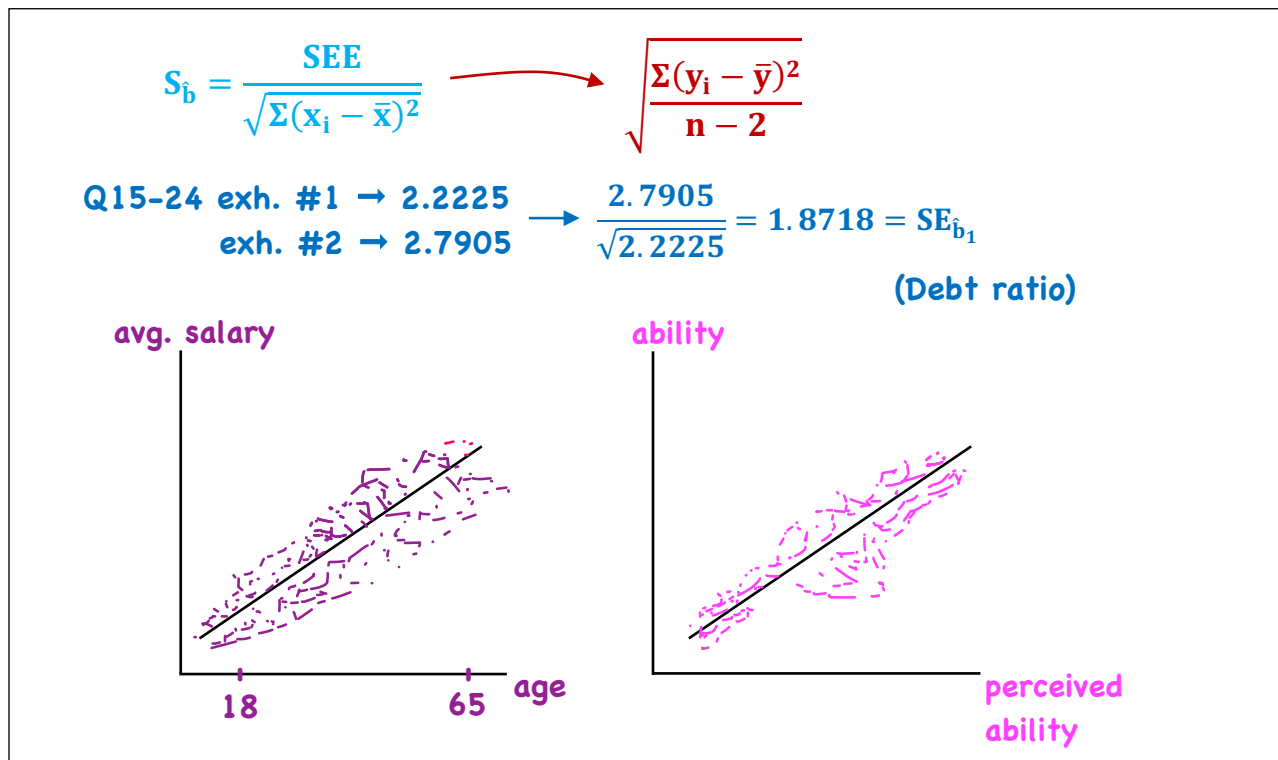
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Standard Errors with CH and Multicollinearity

Conditional Heteroskedasticity



Multicollinearity

$$y = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

$$S_{\hat{b}_1} = \sqrt{\frac{1 - R^2}{(1 - R_{X_k G_k}^2)(N - k - 1)}} \cdot \frac{S_y}{S_x}$$

↓
 $X_1 = b_0 + b_1X_2 + b_2X_3$
↓
calculate R^2

higher correlation between X_1 and X_2 & X_3 , the higher $R_{X_k G_k}^2$, the lower $(1 - R_{X_k G_k}^2)$, the lower the denominator, the larger the SE

⊕ - the smaller b_1 will be (partial coefficient)

⊕ = $b_1 / SE \uparrow$

Time Series Issues (SC, RW, S and CH)

- Time Series is just linear regression

- however, the assumptions of linear regression are usually not satisfied



the DV & IV are not distinct in TS

∴ SC is serious (estimated coefficients will not be consistent)

Step #1: use AR(1) $\Rightarrow$
$$\begin{array}{ccc} X_t & = & b_0 + b_1 X_{t-1} \\ \text{DV} & & \text{IV} \end{array}$$

$$X_t = b_0 + b_1 X_{t-1} + \varepsilon$$

AR(1)

But: new problem

$\Rightarrow$ must be Covariance Stationary

(constant & finite mean & variance)

• first differencing
 $|b_1| < 1$

- for SC & AR(p) models, use autocorrelations of the residual
(standard output from stat. software)

- looking for insignificant t-stats (H_0 : autocorrelations = 0)

Z $\rightarrow$ 1.96

($< \sim 2.0$)

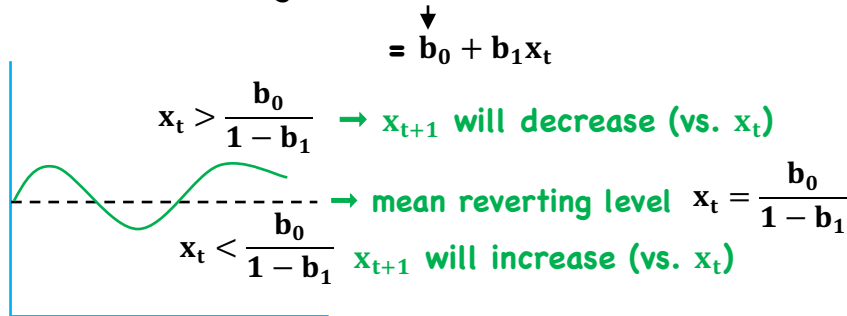
$\alpha = .05$

- no SC.

Lag	Autocorrelation	Standard Error	t-stat.
1	AC_1	$1/\sqrt{T}$	$AC_1/(1/\sqrt{T}) < \text{crit.-t}$
2	AC_2		$AC_2/(1/\sqrt{T}) < \text{crit.-t}$
3			
4			$AC_4/(1/\sqrt{T}) \rightarrow > \text{crit.-t}$
...			
12	AC_{12}		$AC_{12}/(1/\sqrt{T})$

Random Walks/

- first $\Rightarrow$ mean reversion
 - all Cov. St. TS have a mean reverting level
- at a mean reverting level, $x_{t+1} = x_t$



Random Walk $\rightarrow x_{t+1} = x_t + \varepsilon_t$

($b_0 = 0, b_1 = 1$)

- undefined mean reverting level
- no finite variance, $\therefore$ not Cov. St.

ε - constant variance

$\rho_{\varepsilon_t \varepsilon_{t-1}} = 0$

Random Walks/

$\Rightarrow$ We cannot use standard regression analysis on a time series that is a random walk

- first differencing: new series $y_t = \Delta x_t = x_t - x_{t-1}$

RW: $b_0 = 0 \quad b_1 = 0, y_t = \varepsilon_t$

• mean reverting level = $\frac{0}{1 - 0} = 0$

Cov. St. TS

- profit margins
- inflation rates
- Dividend yields

Non-Cov. St. TS

- quarterly sales
- CPI, GDP
- EPS
- prices

must first difference

- if RW:

$b_0 = 0$ } non-sig.

$b_1 = 0$ } t-stats.

no SC in ε_t

if $b_0 \neq 0$ } RW w/ drift.
 $b_1 = 0$

• Example #10

Random Walks/

• Unit Root test of Nonstationarity/

Dickey-Fuller test for a unit root (DF) ($H_0 = \text{unit root}$)

AR(1) $\Rightarrow |b_1| < 1 = \text{Cov. St.}$
(i.e. not first diff.) $|b_1| = 1 = \text{not Cov. St., RW, unit root}$

- issue $\Rightarrow$ if $|b_1| = 1$, then x_t is not Cov. St. & t-stat will not follow a t-distribution
(invalid t-tests)

Seasonality/ - so seriously super simple

- the TS shows regular patterns of movement within the year $\rightarrow$ quarterly (4th Q)
 $\rightarrow$ monthly (12th M) } identified by a sig. autocor. of the error term.
- just use a seasonal lag $\rightarrow X_{t-4}$
 X_{t-12} Example #16

Conditional Heteroskedasticity/ ARCH

variance of error term depends on the value of the IV

(SE will be incorrect)

- typically underestimated

$\varepsilon_t^2 = a_0 + a_1 \varepsilon_{t-1}^2 + \mu_1 \Rightarrow$ estimate this equation
 $\downarrow$ (requires the residuals from a previously estimated regression)
if $a_1 \neq 0$
TS is ARCH(1)

$\Rightarrow$ Example 17

Currency Exchange Rate



\$1.25 → 1.35

(direct)

$CAD/COKE = 1.25$

⇒ What you get for \$1

$COKE/CAD = 1/1.25 = .80$

(indirect)



\$1.99 → 2.09

$CAD/BREAD = 1.99$

$BREAD/CAD = 1/2.09 = .4785$



\$1.24 → 1.25

$CAD/USD = 1.24$

$USD/CAD = 1/1.25 = .80¢ USD$

Exchange Rates/

$P/B \Rightarrow \text{direct } d/f$ e.g. $CAD/USD = 1.2500$ **\$1.25 for a D-coke**

$\Rightarrow \text{indirect } f/d$ $USD/CAD = .8000$ **.8 D-Coke for \$1 CAD**

changes/

$t_0 P/B = 1.25$ $\frac{1.30}{1.25} - 1 = 4\%$ • the B ↑ 4%

$t_1 P/B = 1.30$ $\frac{1/1.30}{1/1.25} - 1 = \frac{.7692}{.80} - 1 = -3.85\%$ • the P ↓ 3.85%

bid-offer/

	<u>B</u>	<u>O</u>
CAD/USD	1.2498	1.2502
USD/CAD	$\frac{1}{1.2502}$	$\frac{1}{1.2498}$
	= .7999	= .8001

Real exchange rate: $S_{d/f}$ CPI_f CPI_d

e.g./ $CAD/USD = 1.2500 \Rightarrow$ for \$1.25 CAD, we get \$1 USD of purchasing power

$\rightarrow$ Price level in US $\uparrow$ 2%

$1.2500 \times 1.02 = 1.275 \Rightarrow$ for \$1.275 CAD, we get \$1 USD of purchasing power

- now assume the CAD price level $\uparrow$ by 1%

$\frac{1.275}{1.01} = 1.2624 \Rightarrow$ for every \$1.2624 CAD, we get \$1 USD of purchasing power

$$\frac{S_{d/f} \times P_f}{P_d} = S_{d/f} \times \left(\frac{P_f}{P_d} \right) \quad \text{or/} \quad \frac{S_{p/B} \times CPI_B}{CPI_p} = S_{p/B} \cdot \left(\frac{CPI_B}{CPI_p} \right) \quad \text{absolute PPP}$$

cross rates $A/B \times B/C = A/C$

e.g./ $CAD/USD = 1.2493$ $USD/EUR = 1.2220$

$$CAD/EUR = CAD/USD \times USD/EUR = 1.2493 \times 1.2220 = 1.5266$$

- if we have A/B & A/C , and we want C/B , must first invert A/C

e.g./ $CAD/USD = 1.2493$ $CAD/JPY = .01115$

$$\begin{aligned} JPY/USD &= CAD/USD \times JPY/CAD \\ &= 1.2493 \times 1/.01115 = 1.2493 \times 89.686 \\ &= 112.04 \end{aligned}$$

- Decompose: CHF/EUR

$$= USD/EUR \times CHF/USD$$

	<u>bid</u>	<u>offer</u>	Step 1 Sell \$1000 USD for EUR
EUR/USD	€1.000	€1.0015	→ Dealer buys USD @ €1.000
GBP/USD	£2.0000	£2.0100	sells USD @ €1.0015
EUR/GBP	€ .3985	€ .4000	1000 x 1.000 = €1000

Arb. opportunity?

$$\text{EUR/USD} \times \text{USD/GBP}$$

$$1.000 \times \frac{1}{2.0100} = .4975 \quad \text{vs.} \quad .3985$$

$$1.0015 \times \frac{1}{2.000} = .50075 \quad \text{should be} \quad .4000 \quad \text{buy low}$$

Step #2 Sell €1000 EUR for GBP = buy GBP

- Dealer buys GBP for € .3985
sells GBP for € .4000

Step #3. Sell GBP for USD (= buy USD)
Dealer buys USD for £2.000
sells USD for £2.010

$$2500 / 2.010 = 1243.78$$

$$€1000 / € .400 = £2500$$

	<u>bid</u>	<u>offer</u>	Step 1 Sell \$1000 USD for GBP
EUR/USD	€1.000	€1.0015	→ Dealer buys USD @ £2.000
GBP/USD	£2.0000	£2.0100	sells USD @ £2.0100
EUR/GBP	€ .5985	€ .6000	1000 x 2.000 = £2,000

Arb. opportunity?

$$\text{EUR/USD} \times \text{USD/GBP}$$

$$1.000 \times \frac{1}{2.0100} = .4975 \quad \text{vs.} \quad .5985 \quad \text{sell high}$$

$$1.0015 \times \frac{1}{2.000} = .50075 \quad \text{should be} \quad .6000 \quad \text{is}$$

Step #2 Sell 2000 GBP for EUR

- Dealer buys GBP for € .5985
sells GBP for € .6000

Step #3. Sell EUR for USD (= buy USD)
Dealer buys USD for €1.0000
sells USD for €1.0015

$$1197 / 1.0015 = 1195.21$$

$$2000 \times .5985 = €1197$$

forward rates/

$$r_{CAD} = 1\%$$

$$r_{USD} = 3\%$$

$$CAD/USD = 1.2500$$

$$USD/CAD = .8000$$

$$\frac{1.2500(1.01)}{1.03} = 1.2257$$

$$\frac{.8000(1.03)}{1.01} = .8158$$

$$1/1.2257 = .8158$$

$$F_{d/f} = S_{d/f} \left(\frac{1 + r_d}{1 + r_f} \right)$$

$$F_{f/d} = S_{f/d} \left(\frac{1 + r_f}{1 + r_d} \right)$$

- I will have USD in the future, locking in a CAD rate

- I will have CAD in the future, locking in a US rate

⇒ quoted: Spot + points → forward premium $(1 + r_p) > (1 + r_B)$

Spot - points → forward discount $(1 + r_p) < (1 + r_B)$

• Bought \$1M CAD for delivery against the USD in 1 yr.

$$? \Rightarrow CAD/USD = .8000/.8004 \quad USD/CAD = 1.2493/1.2500$$

$$1 \text{ yr.} \Rightarrow \text{Forward points } \underline{158.4/162.5} \quad \underline{-242.7/-242.8}$$

$$F_0 = ?$$

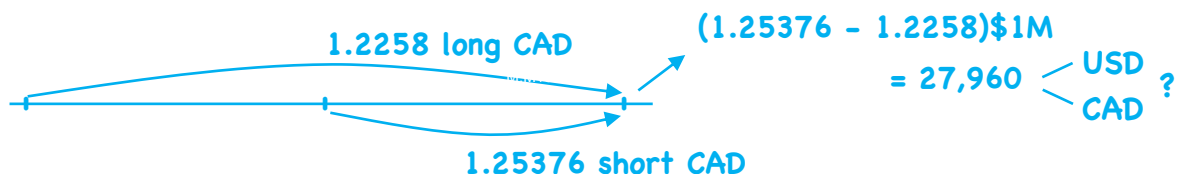
$$6\text{-mos. later} \quad S_{USD/CAD} = 1.2631/1.2634$$

$$r_{CAD} = 2.75\%$$

$$r_{USD} = 1.25\%$$

$$6 \text{ mos.} \Rightarrow \text{f-points } -93.4/-93.5$$

? Sell \$1M CAD
Buy USD



CIRP
|
hedged $F_{f/d} = S_{f/d} \left(\frac{1 + r_f}{1 + r_d} \right)$

UIRP
|
unhedged $\% \Delta S_{f/d}^e \cong (i_f - i_d)$

Forward Rate Parity

- the forward rate will be an unbiased predictor of the future spot fx.-rate

i.e. $F_{f/d} = S_{f/d}^e$

Absolute PPP

$$S_{f/d} = \frac{CPI_f}{CPI_d}$$

Relative PPP

$$\% \Delta S_{f/d} \cong (\pi_f - \pi_d)$$

ex-ante PPP

$$\% \Delta S_{f/d}^e \cong (\pi_f^e - \pi_d^e)$$

$$i_f - i_d = (r_f - r_d) + (\pi_f^e - \pi_d^e)$$

$$(r_f - r_d) = (i_f - i_d) - (\pi_f^e - \pi_d^e)$$

$\downarrow$
0

$\downarrow$
 $\% \Delta S_{f/d}^e$

$\downarrow$
 $\% \Delta S_{f/d}^e$

$$i_f - i_d = \pi_f^e - \pi_d^e$$

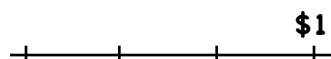
if UIRP + ex-ante hold,
then real int. rate parity holds

Spot, Forward and Par Rates/Curves

1) Determining Treasury Spot Rates

2) Determining Forward Rates

3) Determining Par Rates



1. Spot Rates/ - these are derived from Treasury Bonds

e.g./

Bond FV	TTM	Coupon	PV	Spot	
100	.25	0	97.5	10.256	$\frac{100 - 97.5}{97.5} \times 4 = 10.256$
100	.50	0	94.9	10.748	$\frac{100 - 94.9}{94.9} \times 2 = 10.748$
100	1.00	0	90.0	11.11%	$\frac{100 - 90}{90} = .1111$
100	1.50	8	96.0		
100	2.00	12	101.60		

1. Spot Rates/

e.g./

Bond FV	TTM	Coupon	PV	Spot	
100	.25	0	97.5	10.256	
100	.50	0	94.9	10.748	$\frac{6}{(1.10748)^5} + \frac{6}{(1.1111)}$
100	1.00	0	90.0	11.11%	
100	1.50	8	96.0	11.27%	
100	2.00	12	101.60	11.42%	$\frac{6}{(1.1127)^{1.5}} + \frac{106}{(1+r)^2} = 101.6$
					$= 11.42\%$
					$\frac{4}{(1.10748)^5} + \frac{4}{(1.1111)} + \frac{104}{(1+r)^{1.5}} = 96$
					$\frac{104}{(1+r)^{1.5}} = 88.60$
					$r = \left(\frac{104}{88.60}\right)^{1/1.5} - 1$
					$= .1127$

2. Forward Rates/

Spot

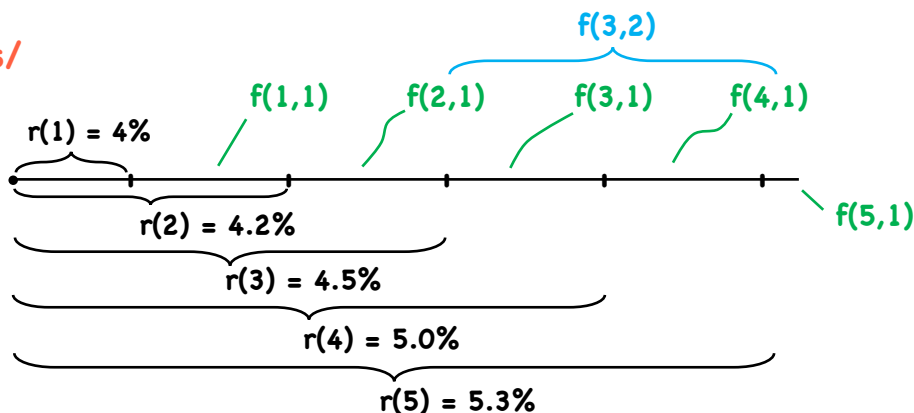
$$r(1) = 4\%$$

$$r(2) = 4.2\%$$

$$r(3) = 4.5\%$$

$$r(4) = 5.0\%$$

$$r(5) = 5.3\%$$



$$(1.042)^2 = (1.04)(1 + r)$$

$$\frac{(1.042)^2}{1.04} = 1 + r$$

$$r = \frac{(1.042)^2}{1.04} - 1$$

$$f(2,1) = \frac{(1.045)^3}{(1.042)^2} - 1$$

$$f(3,2) = \left(\frac{(1.053)^5}{(1.045)^3} \right)^{1/2} - 1$$

$$f(1,4) = \left(\frac{(r(5))^5}{r(1)} \right)^{1/4} - 1$$

2. Forward Rates/

Spot

$$r(1) = 4\%$$

$$r(2) = 4.2\%$$

$$r(3) = 4.5\%$$

$$r(4) = 5.0\%$$

$$r(5) = 5.3\%$$

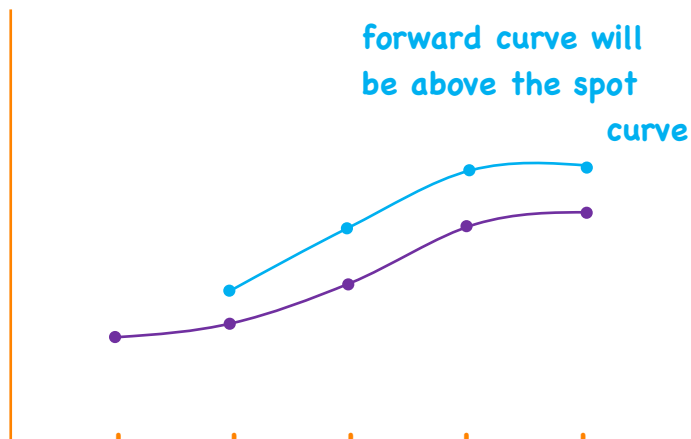
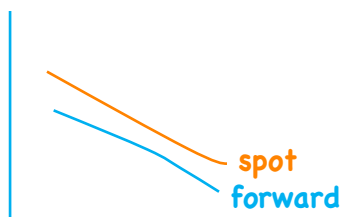
f(x,1)

$$4.40\%$$

$$5.103\%$$

$$6.51\%$$

$$6.51\%$$



- upward sloping spot curve

= upward sloping forward curve

- forward curve will be below spot curve

3/ Par rates/

Spot	Par rate	DF
r(1) = 4%	4%	.96154
r(2) = 4.2%	4.196	.9210
r(3) = 4.5%	4.484	.87629
r(4) = 5.0%	4.95	.8227
r(5) = 5.3%	5.226	.77243

$$PMT_4 = \frac{1 - .8227}{3.58153} = 4.95$$

$$PMT_5 = \frac{1 - .77243}{4.35396} = 5.226$$

$$\frac{PMT + 100}{1.04} = 100 \quad 4\%$$

$$\frac{PMT}{1.04} + \frac{PMT}{(1.042)^2} + \frac{1}{(1.042)^2} = 1$$

$$PMT \left(\frac{1}{1.04} + \frac{1}{(1.042)^2} \right) + \frac{1}{(1.042)^2} = 1$$

$$PMT(\Sigma DF) + DF_2 = 1$$

$$PMT(\Sigma DF) = 1 - DF_2$$

$$PMT = \frac{1 - DF_2}{\Sigma DF}$$

3/ Par rates/

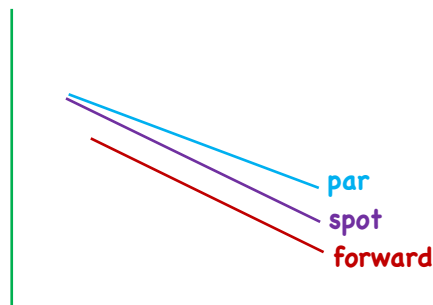
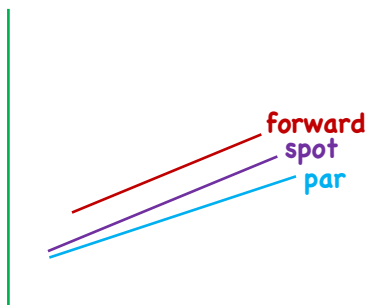
Spot	Par rate	DF
r(1) = 5.3%	5.3%	.94967
r(2) = 5.0%	5.007%	.90703
r(3) = 4.5%	4.526%	.8763
r(4) = 4.2%	4.237%	.84826
r(5) = 4.0%	4.044	.82193

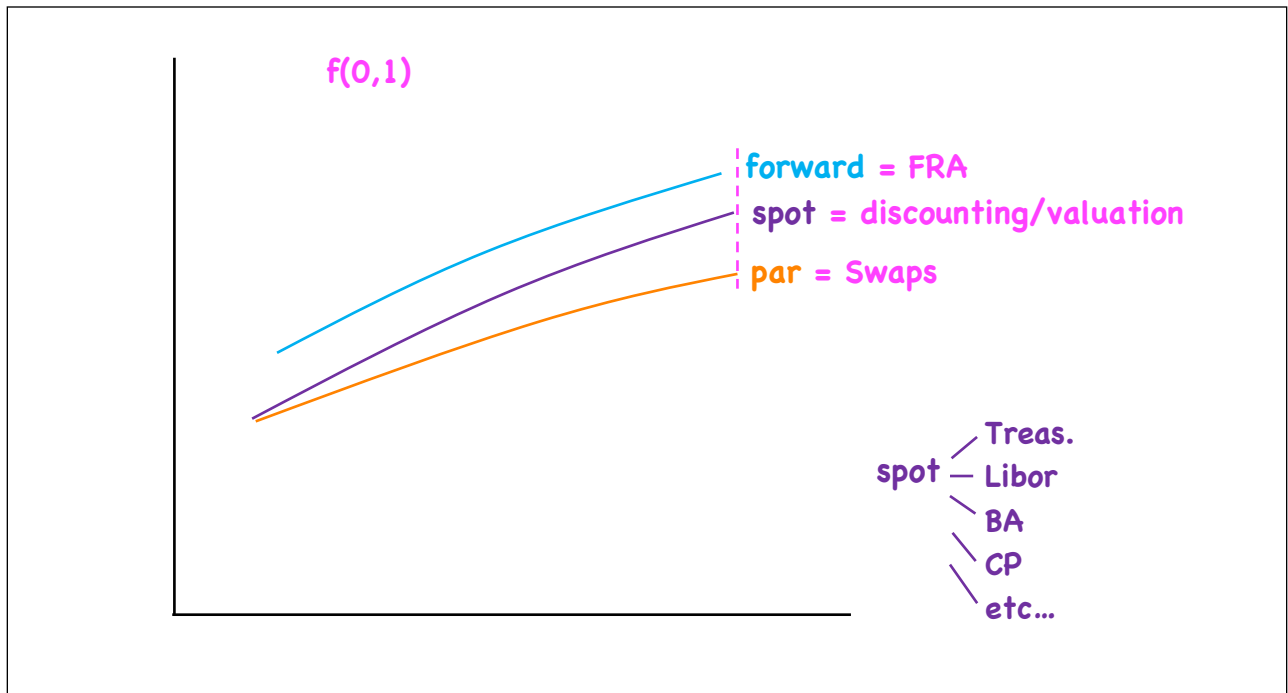
$$PMT_2 = \frac{1 - .90703}{1.8567} = 5.007\%$$

$$PMT_3 = \frac{1 - .8763}{2.733} = 4.526\%$$

$$PMT_4 = \frac{1 - .84826}{3.58126} = 4.237\%$$

$$PMT_5 = \frac{1 - .82193}{4.40319} = 4.044\%$$





Why are ModDur and EffDur different for the same bond?

Par rates

Term	Par rate	Spot
1 yr.	2.2%	2.20%
2 yr.	2.6%	2.605221%
3 yr.	3.0%	3.016357%
4 yr.	3.40%	3.456533%
5 yr.	3.90%	3.975651%

5 yr., 4% annual bond

PV = 100.444356

N = 5, FV = 100, PMT = 4, PV = -100.444356

CPT $I/Y = 3.9004645\%$

-1% +1%

- calculate new spots - calculate new spots

105.095743 96.012392

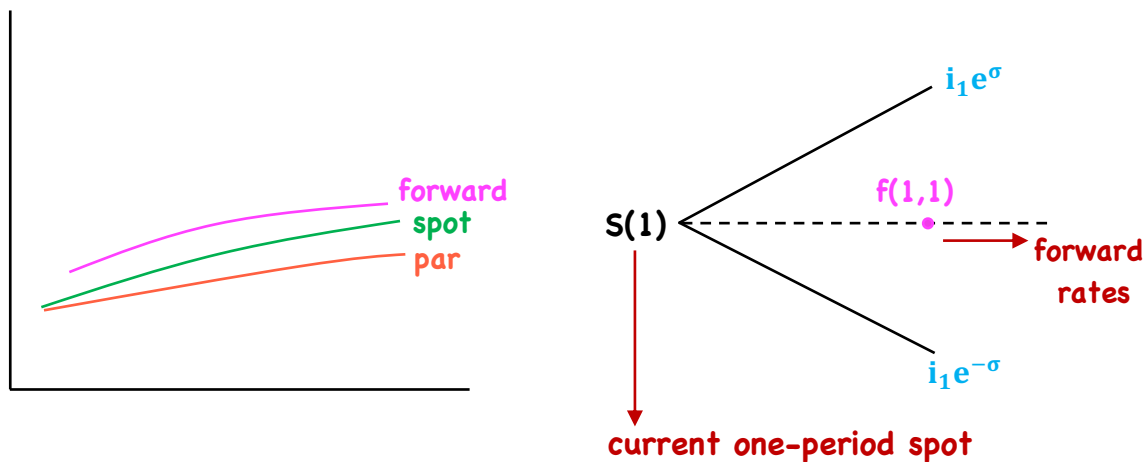
EffDur

$$\frac{PV^- - PV^+}{2 \times \Delta_{\text{curve}} \times PV_0} = 4.52158364$$

ModDur

$$\frac{PV^- - PV^+}{2 \times \Delta_{\text{yield}} \times PV_0} = 4.4597746$$

Interest Rate Trees



- par rates
(constant)

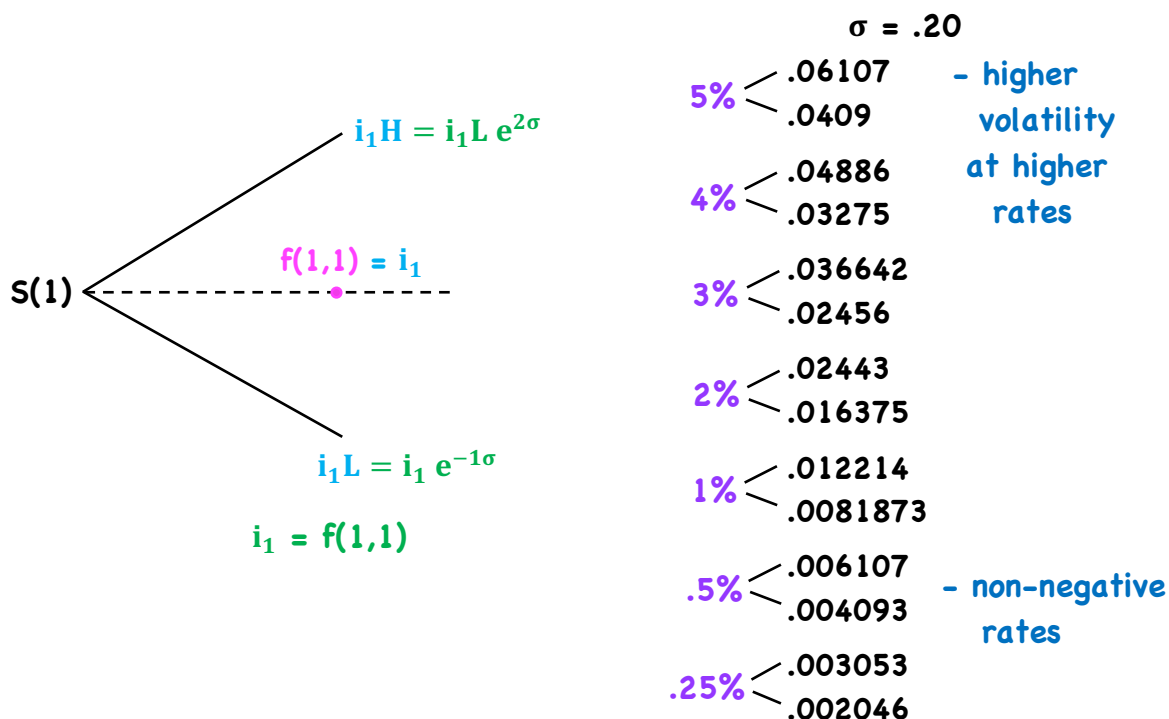
$$\frac{\text{PMT}}{\text{par rate}} + \frac{\text{PMT}}{(\text{par rate})^2} + \dots + \frac{\text{PMT} + \text{FV}}{(\text{par rate})^n} = \underline{100}$$

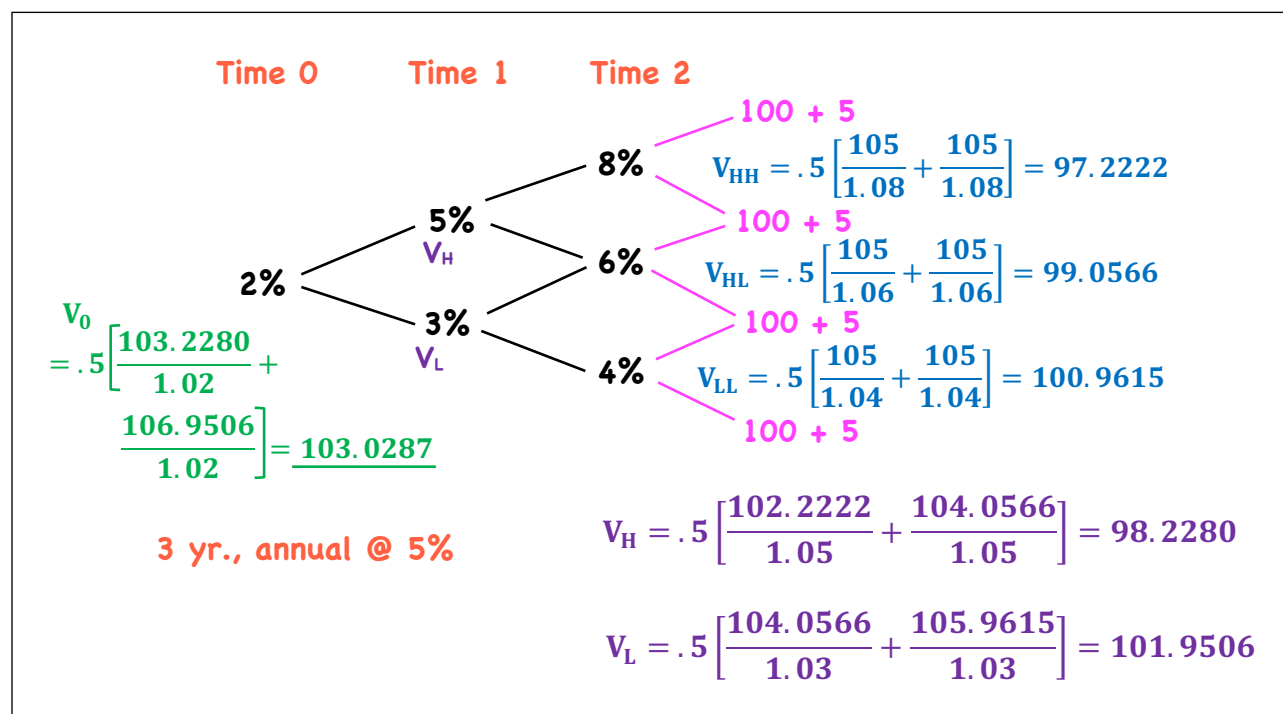
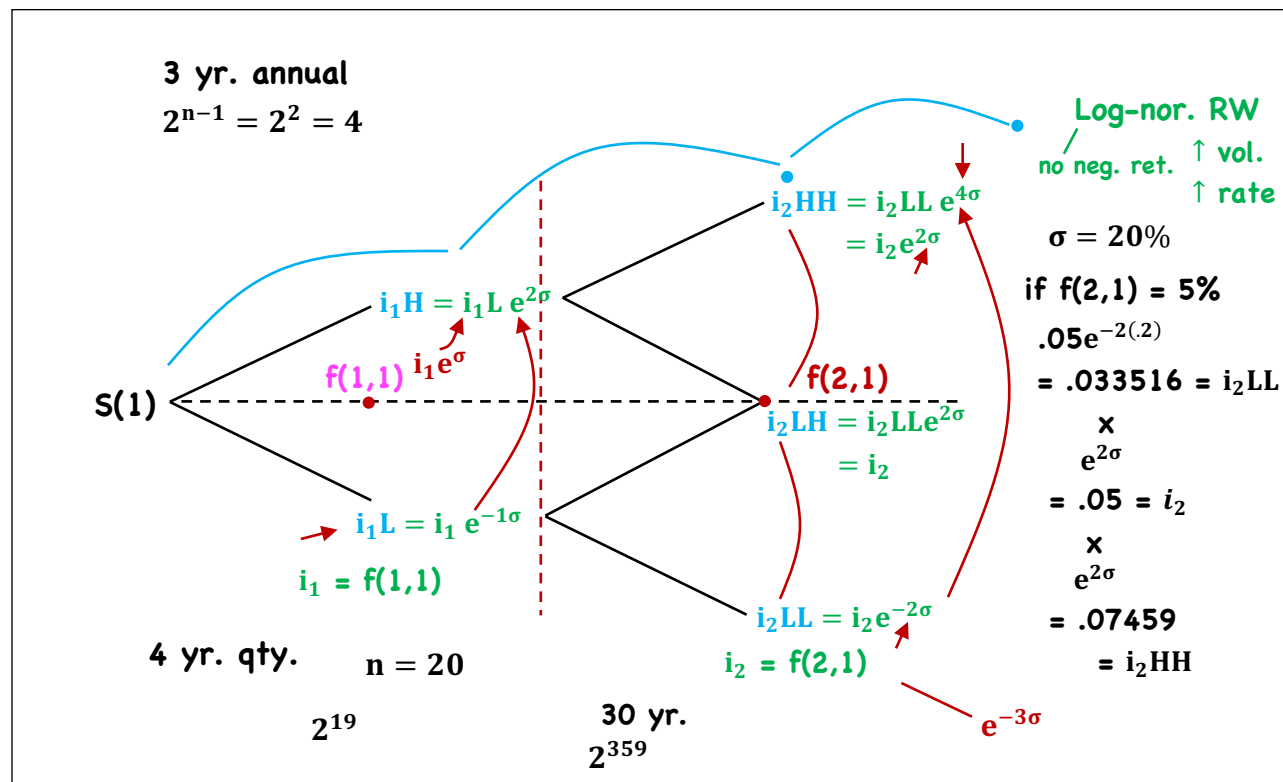
- spot rates

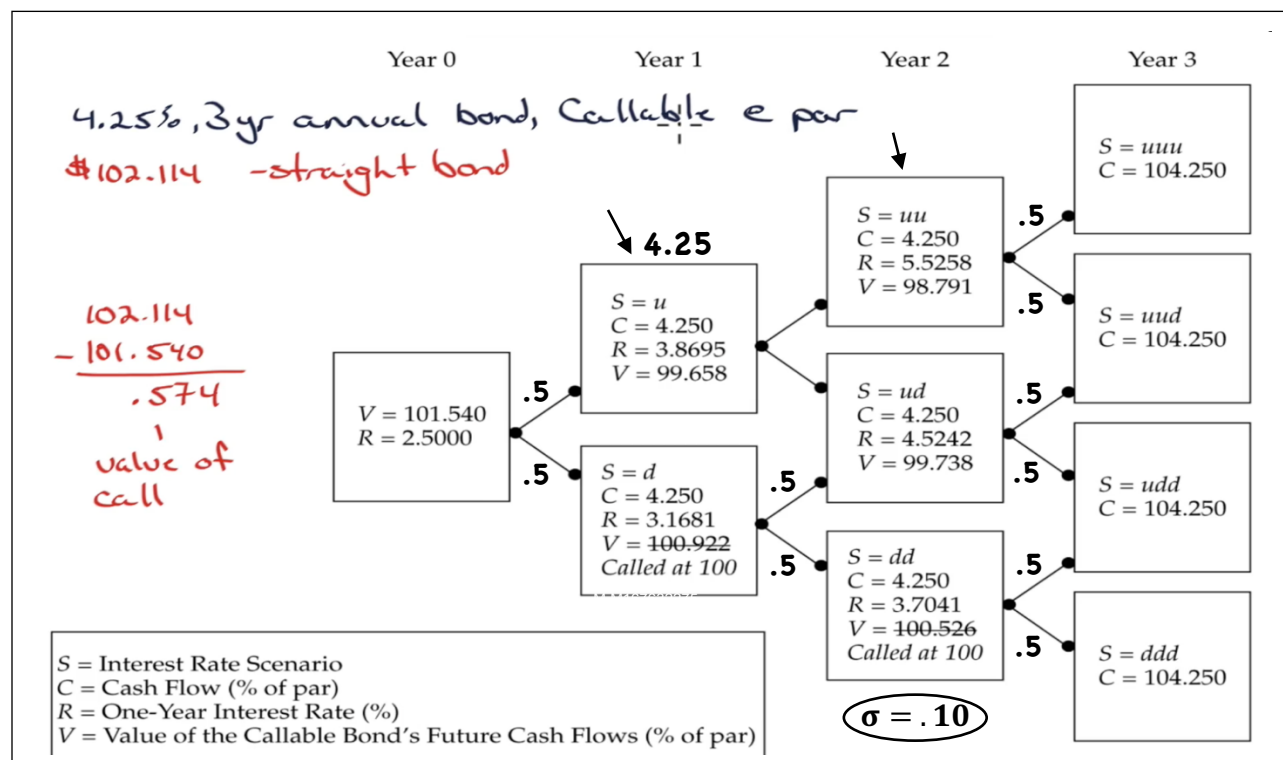
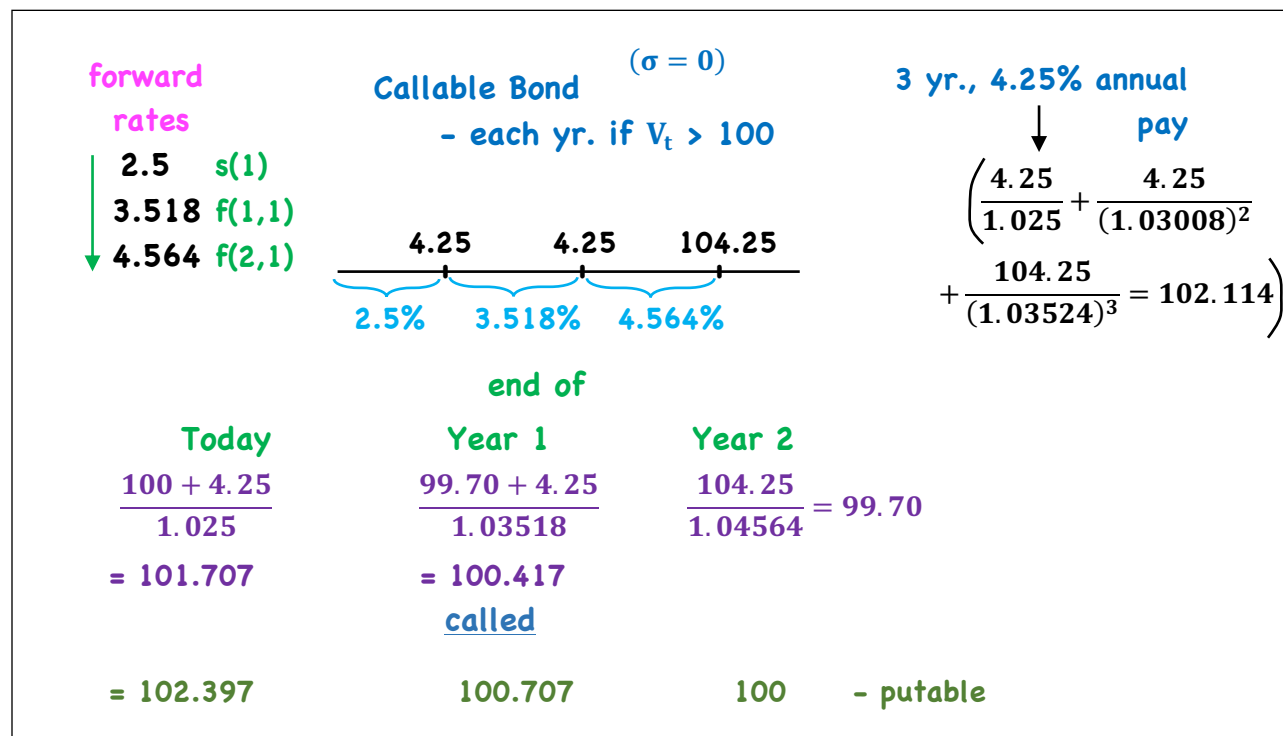
$$\frac{\text{PMT}}{\text{spot}_1} + \frac{\text{PMT}}{(\text{spot}_2)^2} + \dots + \frac{\text{PMT} + \text{FV}}{(\text{spot}_n)^n} = \underline{100}$$

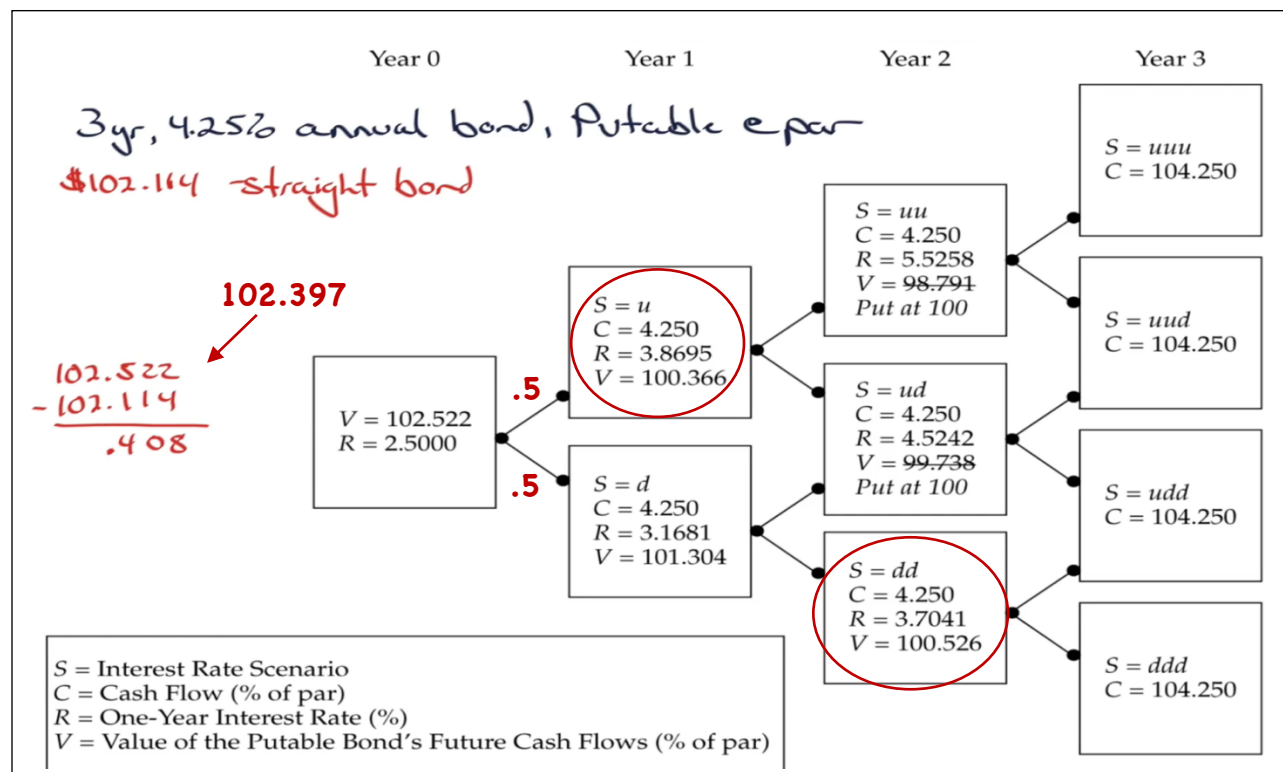
- forward rates

$$\frac{\text{PMT}}{\text{spot}_1} + \frac{\text{PMT}}{\text{spot}_1 \times f(1,1)} + \frac{\text{PMT}}{\text{spot}_1 \times f(1,1) \times f(2,1)} + \dots + \frac{\text{PMT} + \text{FV}}{\text{spot} \times f(1,1) \dots \times f(n-1,1)} = \underline{100}$$



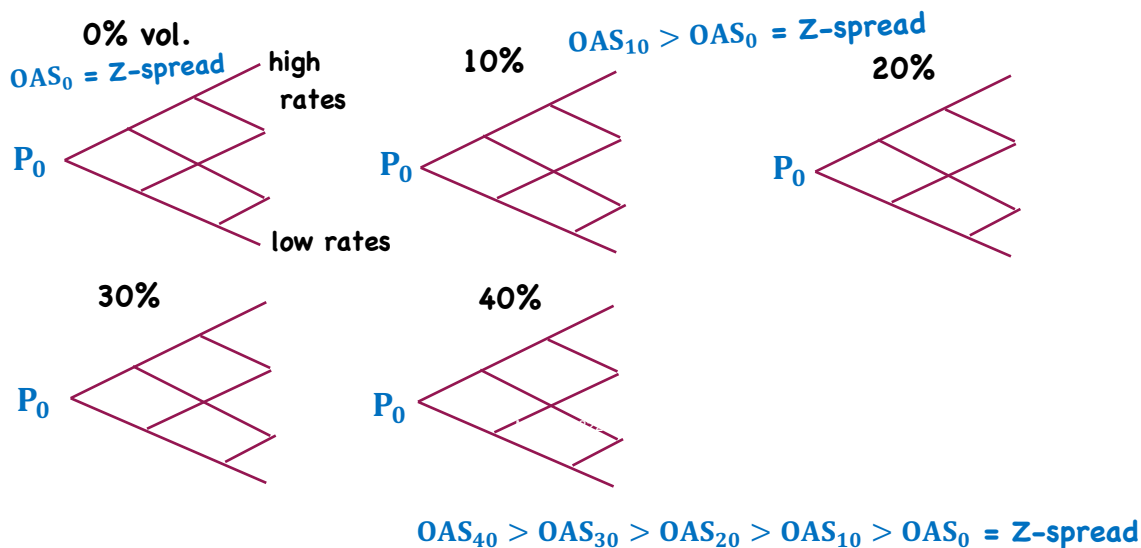
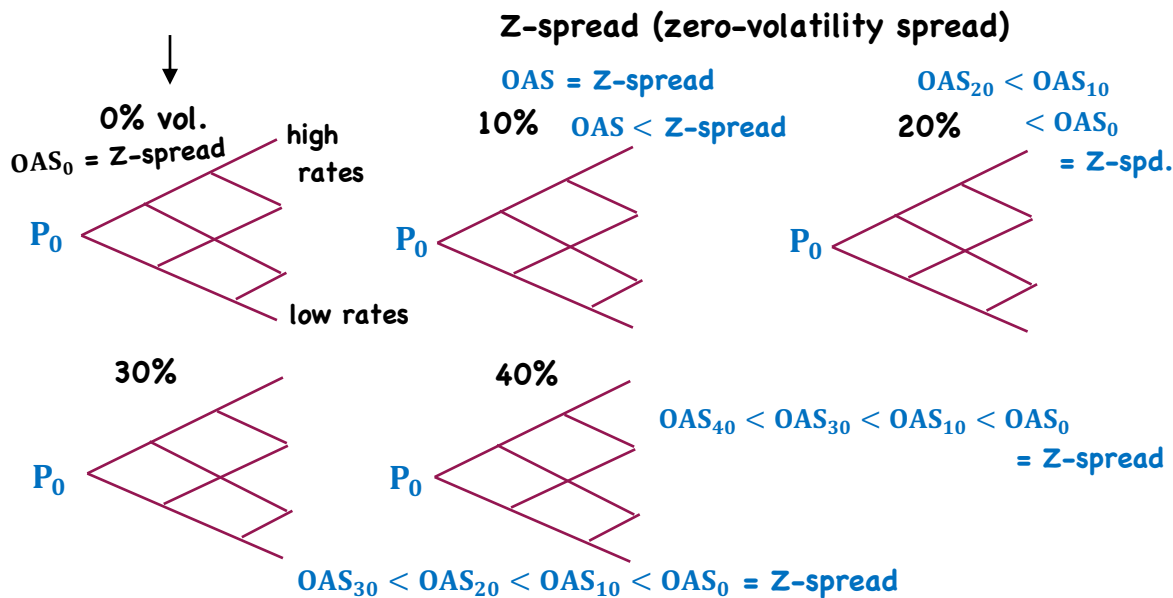






OAS

For a risky bond whose CFs are independent of future interest rates, OAS is essentially same as Z-spread.



$$\text{Z-spread} = OAS_0 + p.$$

$$OAS_0 = OAS_{10} - p$$

Swaps

Interest Rate Swaps

2 legs $\left\{ \begin{array}{l} \text{fixed} - FS_i \\ \text{floating} - S_i \end{array} \right.$

$NA = \$1$

fl. $\swarrow$
rec. fl., pay fx.

Libor $\swarrow$
- advanced set, settled in arrears

$S_1 - FS_0$ $S_2 - FS_0$ $S_3 - FS_0$ $S_4 - FS_0$ $S_5 - FS_0$

1 2 3 4 5

$L(1)$ $L(2)$ $L(3)$ $L(4)$ $L(5)$

long a floating rate bond
Short a fx. rate bond

- for $V_0 = 0$

$V_{fl} = V_{fx}$
 $100 = 100$

$$1 = \frac{PMT}{L(1)} + \frac{PMT}{L(2)^2} + \frac{PMT}{L(3)^3} + \frac{PMT}{L(4)^4} + \frac{PMT + FV}{L(5)^5}$$

$$PMT = \frac{1 - PVF_n}{\Sigma PVF}$$

fx.-rate

Interest Rate Swaps

$$1 = \frac{PMT}{L(1)} + \frac{PMT}{L(2)^2} + \frac{PMT}{L(3)^3} + \frac{PMT}{L(4)^4} + \frac{PMT + FV}{L(5)^5}$$

$$\frac{PMT}{L(5)^5} + \frac{1}{L(5)^5}$$

$$1 = PMT \left(\frac{1}{L(1)} + \frac{2}{L(2)^2} + \dots + \frac{1}{L(5)^5} \right) + \frac{1}{L(5)^5}$$

$$1 = PMT(PVF_1 + PVF_2 + \dots + PVF_5) + PVF_5$$

$$1 - PVF_5 = PMT \left(\sum_{i=1}^5 PVF \right) \quad \text{e.g./}$$

$$PMT = \frac{1 - PVF_n}{\sum_{i=1}^n PVF}$$

YR.

1	1%	$= 1/(1.01) = PVF_1$
2	2%	$= 1/(1.02)^2 = PVF_2$
3	3%	
4	4%	
5	5%	$= 1/(1.05)^5 = \frac{PVF_5}{\Sigma PVF}$

Interest Rate Swaps

e.g./

Maturity	PVF
1	0.99099
2	0.977876
3	0.965136
4	0.951529
5	0.937467
	<u>4.822107</u>

$FS_0 = \text{fixed rate} = ?$

$$FS_0 = \frac{1 - PVF_n}{\Sigma PVF}$$

$$= \frac{1 - .937467}{4.822107} = 1.2968\%$$

$V_{\text{swap}} = 0$

$V_{\text{fx}} = V_{\text{fl}}$

EOC 8/

Maturity	PVF
1	.990099
2	.977876
3	.965136
	<u>2.933111</u>

$$FS_0 = \frac{1 - PVF_3}{\Sigma PVF}$$

$$= \frac{1 - .965136}{2.933111} = .0118864$$

~1.19%

Interest Rate Swaps

Valuation/

rec. fix

pay fl.

$V_0 = \$0$



price out the fixed rate
- enter into an offsetting
rec. fl. pay fx.

rec. fx.

pay fx.

FS_0 - pay fx. rec. fl.
 FS_t rec. fx. pay fl.

$(FS_t - FS_0)$

$V_t = ?$

$$FS_0 - S_3 + S_1 - FS_t \quad FS_0 - S_4 + S_2 - FS_t$$

$$FS_0 - FS_t$$

$$FS_0 - FS_t$$

$$\frac{FS_0 - FS_t}{L(1)} + \frac{FS_0 - FS_t}{L(2)^2}$$

$$FS_0 - FS_t \left(\frac{1}{L(1)} + \frac{1}{L(2)^2} \right)$$

$$NA \cdot FS_0 - FS_t (\Sigma PVF)$$

Interest Rate Swaps

- 2 yrs. ago $\Rightarrow$ entered into a 7 yr. \$100M rec. fx., pay fl. swap

$$FS_0 = 2\%$$

e.g./

Maturity PVF

1 0.99099

2 0.977876

3 0.965136

4 0.951529

5 0.937467

4.822107

$$\frac{1 - PVF_5}{\Sigma PVF}$$

$$= \frac{1 - .937467}{4.822107} = 1.2968\%$$

FS_0
- pay fx, rec fl.

FS_t

$$NA(FS_0 - FS_t)\Sigma PVF$$

$$V_t = 100M(.02 - .013) 4.822107 = \$3,375,000$$

$$(FS_0 - FS_t)$$

Interest Rate Swaps

$$\text{fx rate} = \frac{1 - PVF_n}{\Sigma PVF}$$

$$V_0 = \$0$$

original swap rec. fx., pay fl. $V_t = NA(FS_0 - FS_t)(\Sigma PVF)$

offset $\longrightarrow$ pay fx., rec. fl.

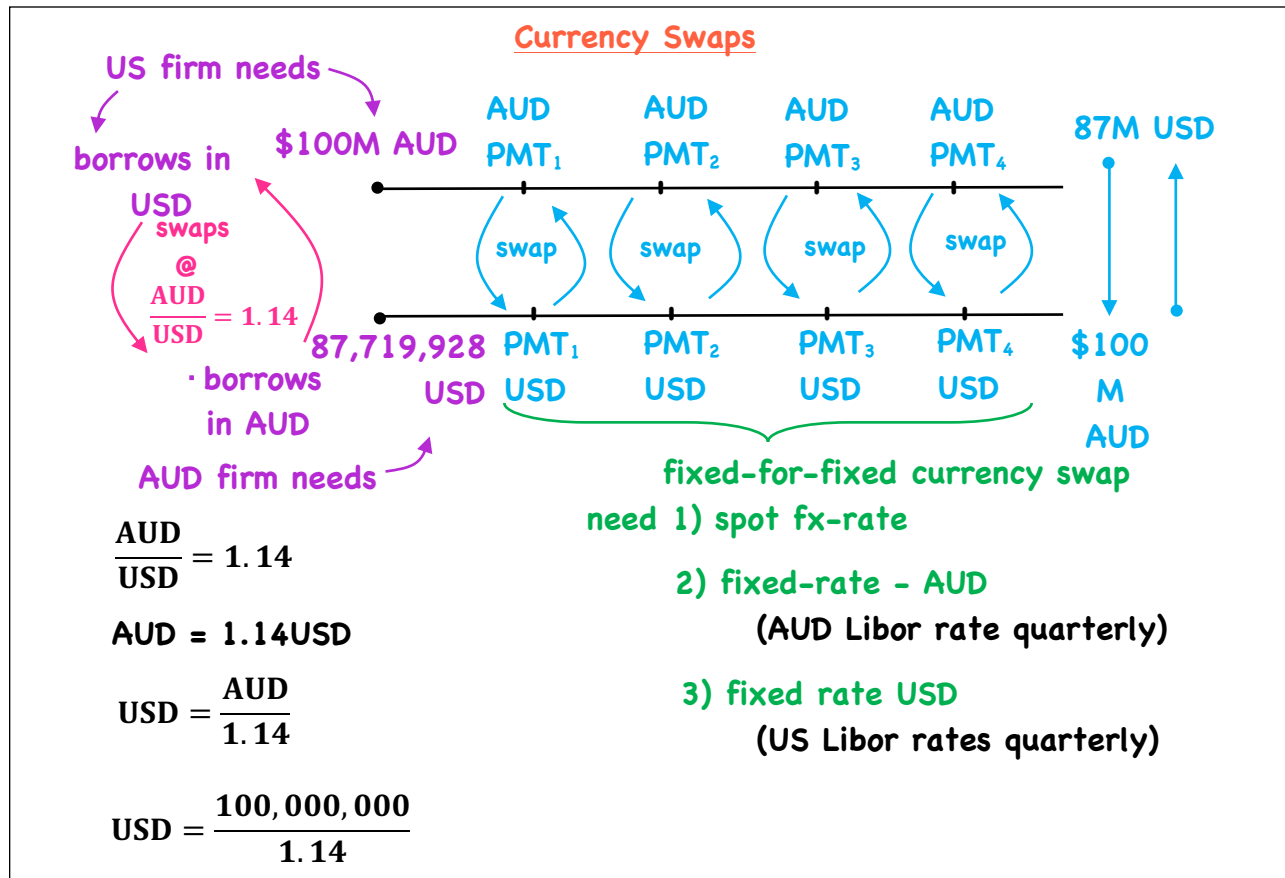
original swap pay fx., rec. fl. or $= NA(FS_t - FS_0)(\Sigma PVF)$

offset $\longrightarrow$ rec. fx., pay fl.

rec. pay

rec. pay

that's it!



Currency Swaps

Days to Maturity	L(AUD)	PVF
90	2.5%	$\frac{1}{(1 + .025(90/360))} = .993789$
180	2.6	$\frac{1}{(1 + .026(180/360))} = .98716$
270	2.7	$\frac{1}{(1 + .027(270/360))} = .980152$
360	2.8	$\frac{1}{(1 + .028)} = .972763$
	L(USD)	= 3.933870
90	.10%	$\frac{1}{(1 + .001(90/360))} = .999750$
180	.15%	$\frac{1}{(1 + .0015(180/360))} = .999251$
270	.20%	$\frac{1}{(1 + .002(270/360))} = .998502$
360	.25%	$\frac{1}{(1 + .0025(1))} = .997506$
		3.995009

$r_{\text{AUD}} = \frac{1 - \text{PVF}_n}{\Sigma \text{PVF}}$
 $= \frac{1 - .972763}{3.933870}$
 $= .00692381$
 $(\times 4 = 2.769524)$

$r_{\text{USD}} = \frac{1 - \text{PVF}_n}{\Sigma \text{PVF}}$
 $= \frac{1 - .997506}{3.995009}$
 $= .00062422$

Currency Swaps

AVD

$$\frac{.00692381}{(L_{90})} + \frac{.00692381}{L_{180}} + \frac{.00692381}{L_{270}} + \frac{.00692381}{L_{360}} + \frac{1}{L_{360}}$$

$$.00692381 \left(\frac{1}{L_{90}} + \frac{1}{L_{180}} + \frac{1}{L_{270}} + \frac{1}{L_{360}} \right) + \frac{1}{L_{360}}$$

$$\$100M[.00692381(\Sigma PVF) + PVF_n] =$$

$\underbrace{3,933.870 \quad .972763}$

100,000,036.80 ~ 100M

$$(1.14)87,719,218[.00062422(3.995009) + .997506] = 99,996,884.97 \sim 100M$$

$$V_{\text{swap}} = \$0$$

Currency Swaps

$$100M[.00692381(\Sigma PVF) + PVF_4] \Rightarrow A$$

NA fixed rate PVF_{AUD}

$$1.14 \times 87,719,218[.00062422(\Sigma PVF) + PVF_4] \Rightarrow B$$

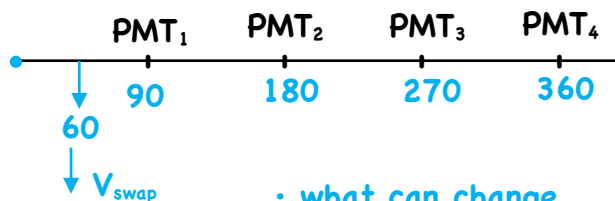
fixed rate PVF_{AUD}

- pay AUD, rec USD

- A + B

- pay USD, rec AUD

A - B

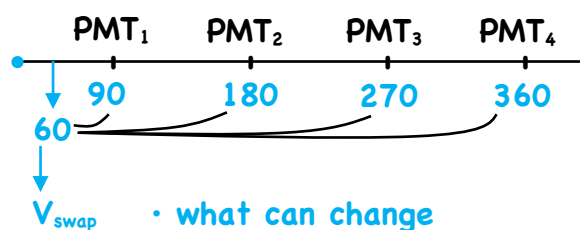


• what can change

① AUD/USD

② L_{AUD} + L_{USD}

Currency Swaps



• what can change

① AUD/USD

② $L_{AUD} + L_{USD}$

	PVF		PVF
AUD	↓	USD	↓
L_{30}	-	L_{30}	-
L_{120}	-	L_{120}	-
L_{210}	-	L_{210}	-
L_{300}	-	L_{300}	-
ΣPVF		ΣPVF	

$\frac{AUD}{USD}$

$$100M[.00692381(3.93387) + .972763] + 1.14 \times 87,719,218[.00062422(3.995009) + .997506]$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 3.967683 .986031 1.13 3.994841 .998336

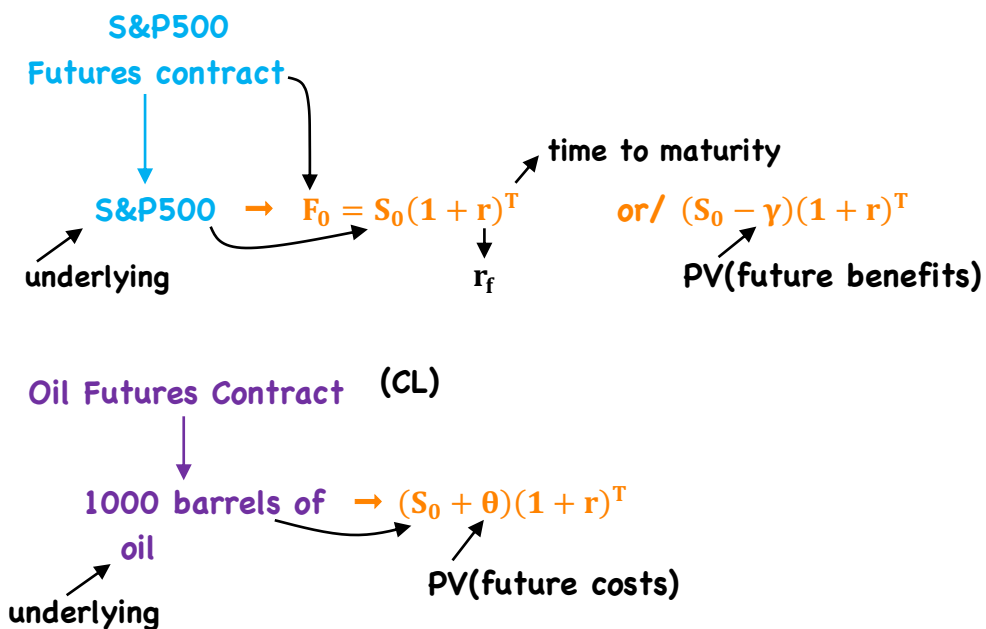
-101,350,248.30 + 99,204,954.46 = -2145293.84

Q: Current V_{swap} to the US company - pay AUD

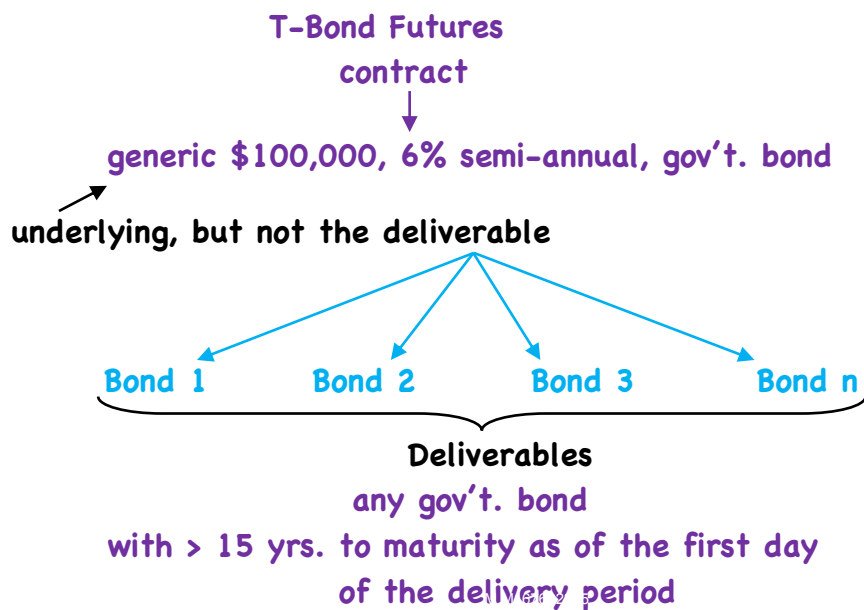
int. forex

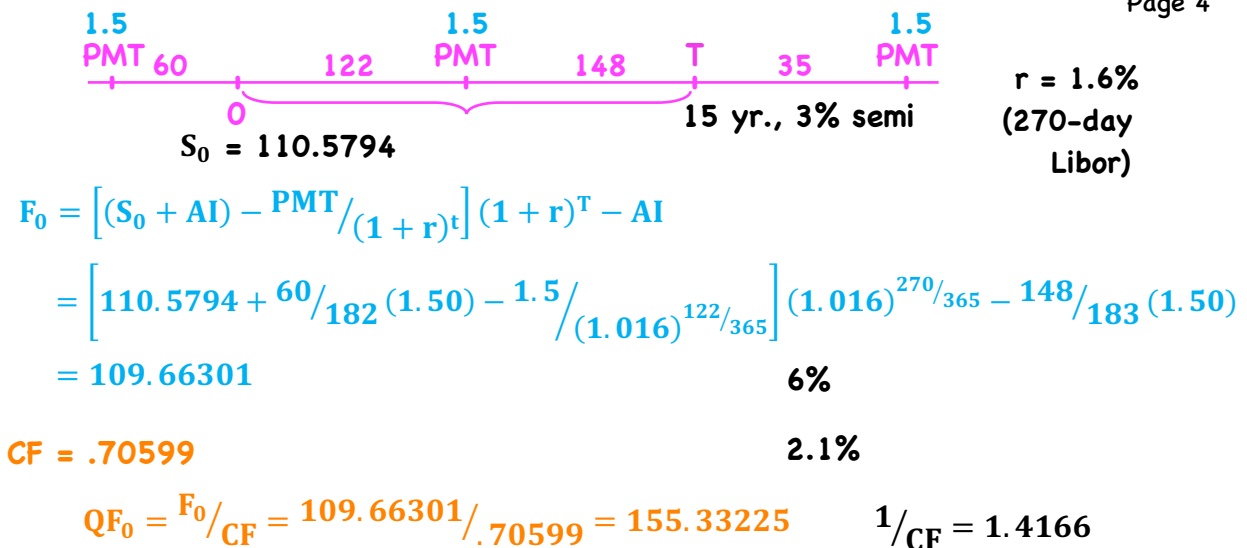
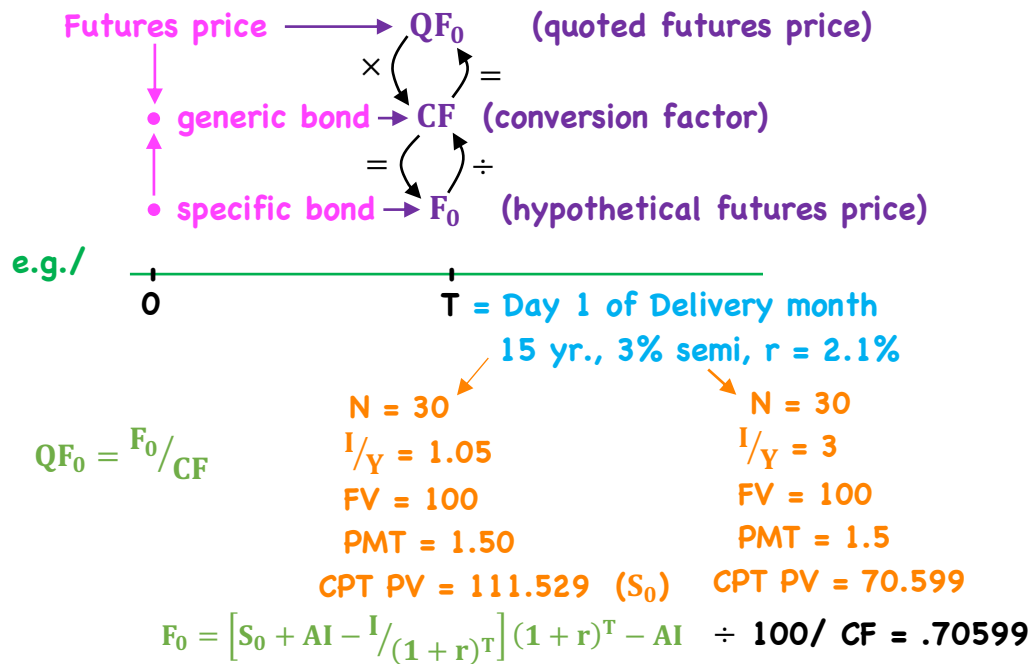
Treasury Bond Futures

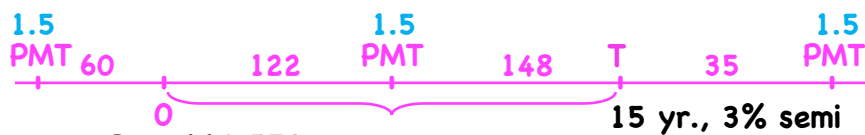
Page 1



Page 2







$$S_0 = 110.5794$$

$r = 1.6\%$
(270-day
Libor)

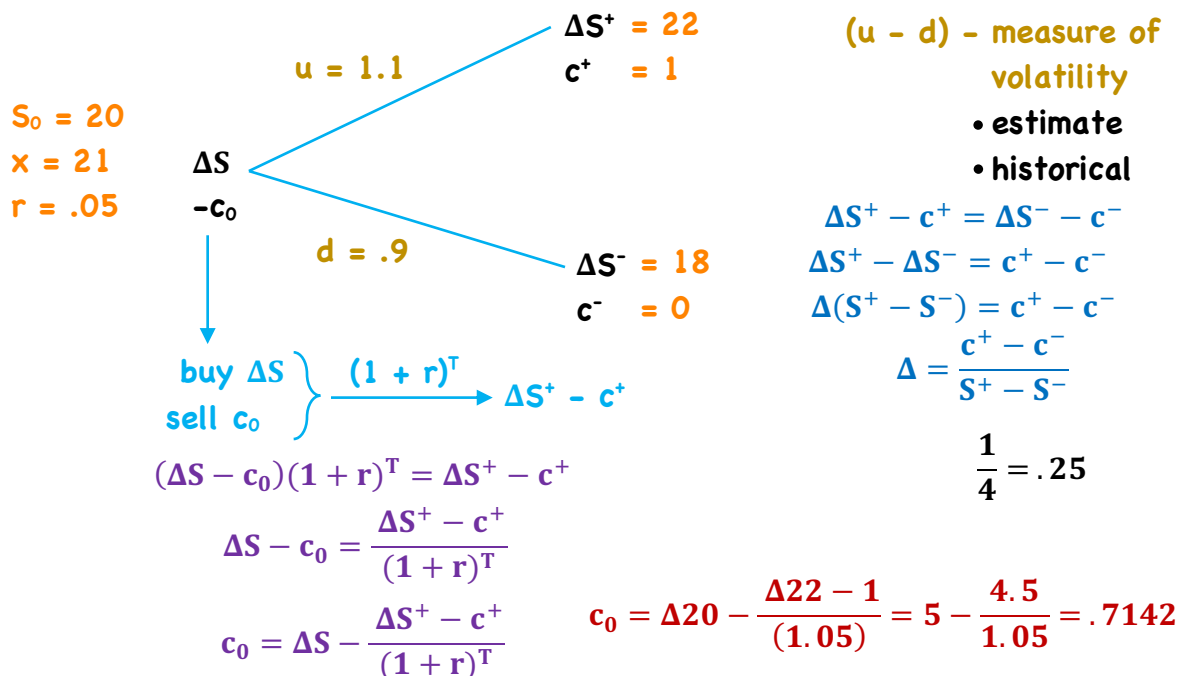
$$QF_0 = \frac{[(S_0 + AI) - \text{PMT}/(1+r)^t] (1+r)^T - AI}{CF}$$

$$QF_0 = F_0 / CF$$

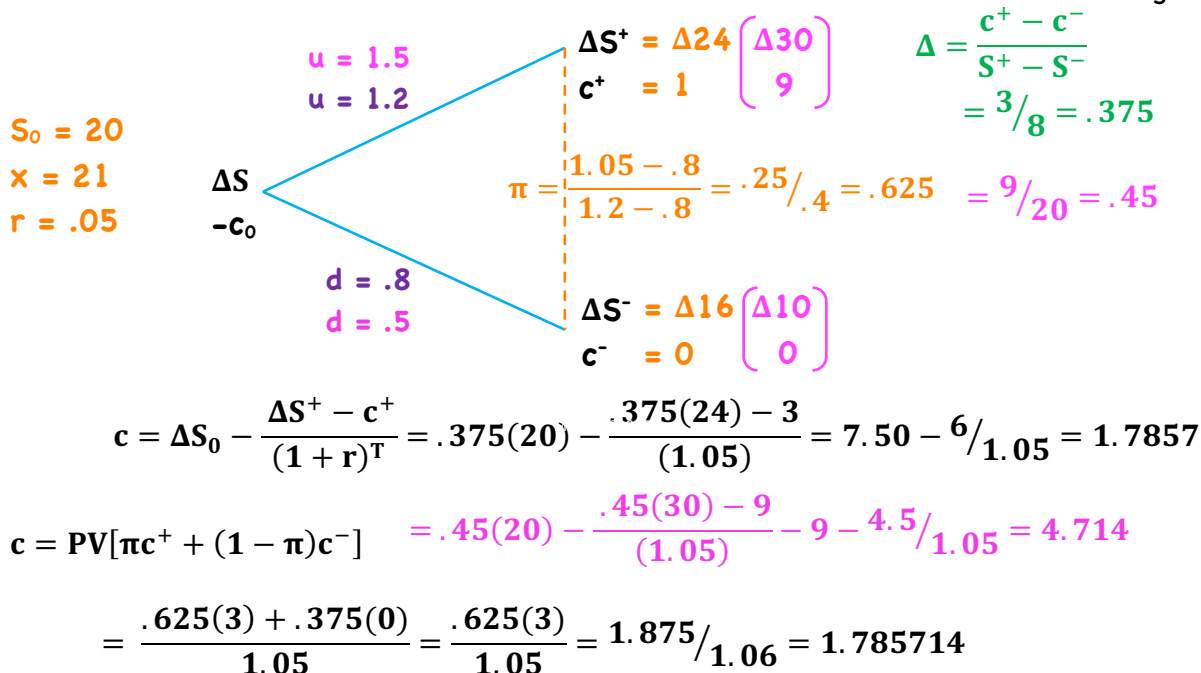
Binomial Option Pricing

Binomial Option Valuation Model

Page 1



Page 2



$S_0 = 67$
 $x = 70$
 $r = 3.1\%$

ΔS
 $-c_0$

$u = 1.27$
 $d = .82$

ΔS^+
 c^+
 ΔS^-
 c^-

$c = PV[\pi c^+ + (1 - \pi)c^-]$
 $p = PV[\pi p^+ + (1 - \pi)p^-]$
 $\pi = \frac{(1 + r) - d}{u - d}$
 same for put or call
 - no need to calculate Δ

$\pi = \frac{1.031 - .82}{1.27 - .82} = .468$
 $c = \frac{.468[(67 \times 1.27) - 70]}{1.031} = \frac{.468(15.09)}{1.031} = 6.8627$
 $p = \frac{(1 - .468)[70 - (67 \times .82)]}{1.031} = 7.758$

$S + p = \frac{x}{(1 + r)^T} + c$
 $67 + p = 70 / (1.031) + 6.8627$
 $p = 70 / (1.031) + 6.8627 - 67$
 $p = 7.7579$

$\frac{c^+ - c^-}{S^+ - S^-}$

s
 c

u
 d

s^+
 c^+
 s^-
 c^-

$\Delta = \frac{c^+ - c^-}{s^+ - s^-}$
 $\Delta^+ = \frac{c^{++} - c^{+-}}{s^{++} - s^{+-}}$
 $\Delta^- = \frac{c^{+-} - c^{--}}{s^{+-} - s^{--}}$

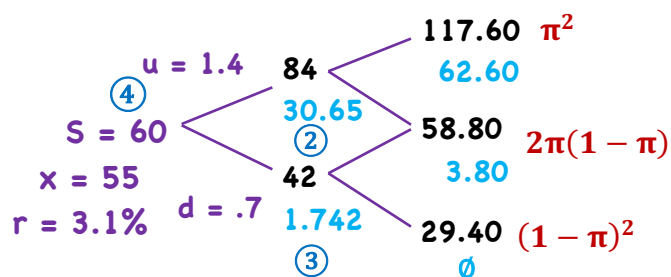
$u = 1.4$
 $d = .7$

$S = 60$
 $x = 55$
 $r = 3.1\%$

84
 30.65
 42
 1.742

117.60
 62.60
 58.80
 3.80
 29.40
 $\emptyset$

$\pi = \frac{1.031 - .7}{1.4 - .7} = .47285$
 $c^+ = \frac{.47285(62.60) + .52714(3.80)}{1.031} = 30.65$
 $c^- = \frac{.47285(3.80)}{1.031} = 1.742$
 $c = \frac{.47285(30.65) + .52714(1.742)}{1.031} = 14.92$

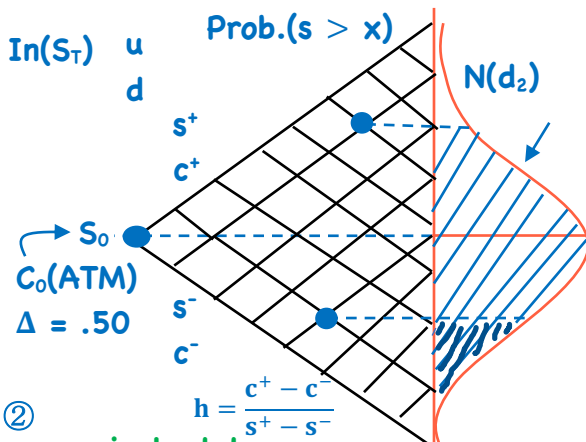


$$\begin{aligned} \textcircled{1} \pi &= \frac{1.031 - .7}{1.4 - .7} = .47285 \\ \textcircled{1} c^+ &= \frac{.47285(62.60) + .52714(3.80)}{1.031} \\ \textcircled{2} &= 30.65 \\ \textcircled{3} c^- &= \frac{.47285(3.80)}{1.031} = 1.742 \\ \textcircled{4} c &= \frac{.47285(30.65) + .52714(1.742)}{1.031} \end{aligned}$$

$$\begin{aligned} \textcircled{2} c &= \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^T} \\ &= \frac{(.47285)^2(62.60) + 2(.47285)(.52714)(3.80)}{(1.031)^2} = 14.92 \\ &= 14.94 \end{aligned}$$

Black-Scholes-Merton (BSM) Model

BSM - no arbitrage approach but in continuous time



② $h = \frac{c}{s^+ - s^-}$
- equivalent to
a binomial process
in which length of each
step approaches zero

- ① - with a large # of steps, binomial process converges to a normal distribution

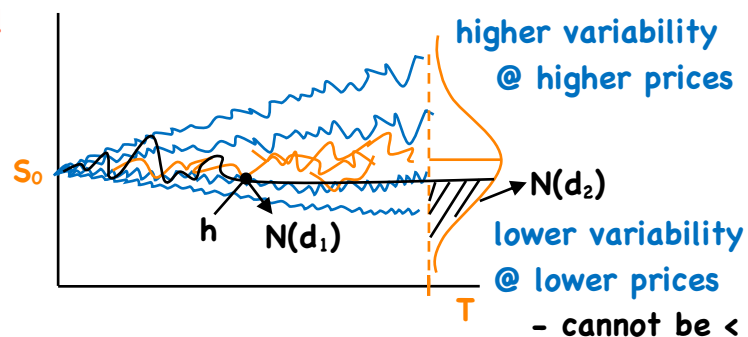
discrete - continuous

Binomial

BSM – Geometric Brownian motion

e.g. ABC $E(R) = 3\%$
 $\sigma = 45\%$

- log-normal



Assumptions/ r_f • is contin. compounded
- known & constant

σ • is known & constant

yield of underlying • contin. compounded, known & constant yield, annualized

contin. compounding since BSM is a continuous time version of the discrete time binomial model

hedger.

$$c = SN(d_1) - e^{rT} \times N(d_2) \quad \text{Prob.}(s_T > x) \quad d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$p = e^{-rT} \times N(-d_2) - SN(-d_1) \quad d_2 = d_1 - \sigma\sqrt{T}$$

Z-values
 $N(d_1)$ $N(d_2)$
 hedge $s > x$

$s = 70$ ($c = s_T - x$)
 $x = 50$ ($p = x - s_T$)

standard normal CUMULATIVE distribution function

this thing

called a normal probability density function

$y = 2x \quad \forall x, y > 0$
 $y = \text{non linear}$

$c = SN(d_1) - e^{rT} \times N(d_2)$ $p = e^{-rT} \times N(-d_2) - SN(-d_1)$ $d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$ $d_2 = d_1 - \sigma\sqrt{T}$					
$S_0 = 50$	$x = 50$	$\sigma = 30\%$	$r = 2\%$	1-YR. call	Call
S	$N(d_1)$	$N(d_2)$	C	P	
25	.01814	.00834	.04486	24.05479	.4535
30	.06863	.03704	.24339	19.25332	.40874
35	.16546	.10164	.80973	14.81967	.04476
40	.29905	.20408	1.96004	10.96997	$N(d_1)$
45	.44649	.33195	3.82319	7.83313	buy .01814
50	.58577	.46679	6.41079	5.42072	shares @ \$25
55	.70346	.59265	9.64435	3.65429	- finance with
60	.79515	.70000	13.40169	2.41162	borrowing of
65	.86241	.78559	17.55496	1.56489	.40874 $N(d_2)$
70	.90959	.85042	21.99229	1.00223	• borrowing less
75	.94158	.89764	26.62559	0.63552	than underlying

$$c = SN(d_1) - e^{rT} \times N(d_2)$$

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$p = e^{-rT} \times N(-d_2) - SN(-d_1)$$

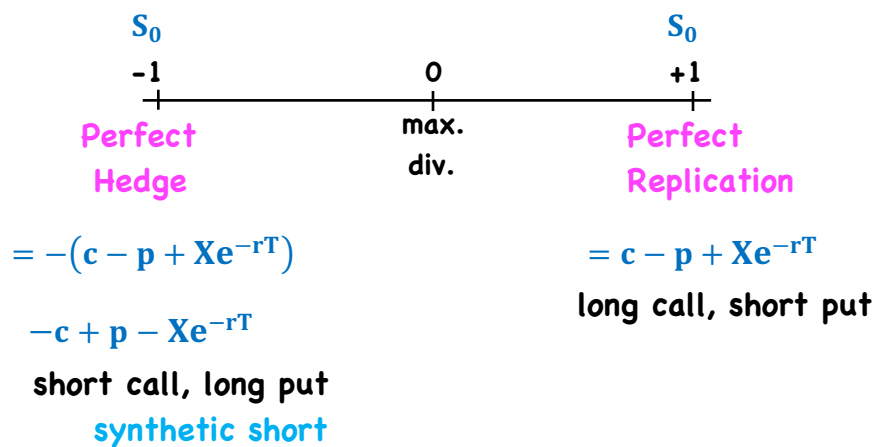
$$d_2 = d_1 - \sigma\sqrt{T}$$

$S_0 = 50$	$x = 50$	$\sigma = 30\%$	$r = 2\%$	1-YR. call	Put
S	$N(d_1)$	$N(d_2)$	C	P	
$s = 25$	$(1 - .01814)$	$(1 - .00834)$	.04486	(24.05479)	Sell \$24.5465
30	.06863	.03704	.24339	19.25332	Invest <u>48.6011</u>
35	.16546	.10164	.80973	14.81967	<u>24.0546</u>
40	.29905	.20408	1.96004	10.96997	Sell $(1 - .01814)$
45	.44649	.33195	3.82319	7.83313	shares @ \$25
$x = 50$	.58577	.46679	6.41079	5.42072	- finance with
55	.70346	.59265	9.64435	3.65429	lending of
60	.79515	.70000	13.40169	2.41162	$(1 - .00834) \times e^{-rT}$
65	.86241	.78559	17.55496	1.56489	• investing more
70	.90959	.85042	21.99229	1.00223	than underlying
75	.94158	.89764	26.62559	0.63552	

Put-Call parity with Replication and Hedging

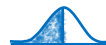
$$S_0 + p = c + Xe^{-rT} \quad - \text{ put-call parity}$$

$$S_0 = c - p + Xe^{-rT} \quad - \text{ synthetic long}$$



Deriving $N(d_1)$

$$Z = \frac{X - \mu}{\sigma} \rightarrow d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$N(d_1)$ 

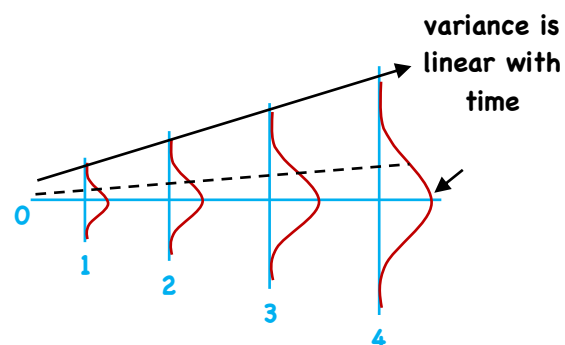
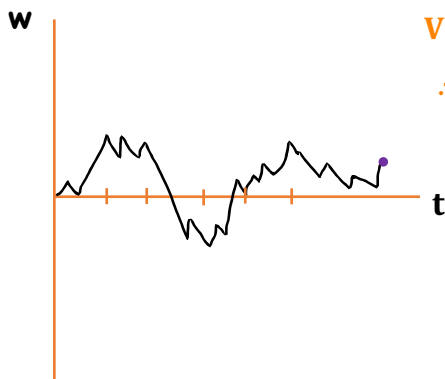
from here  to here

Step 1: Brownian Motion

$$E(W_t) = 0 \quad \text{RW}$$

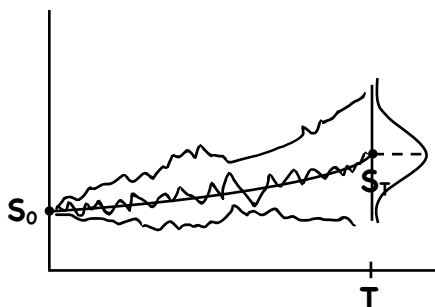
$$\text{Var}(W_t) = t \rightarrow \text{sd} = \sqrt{t}$$

$$\therefore W(t) \sim N(0, t)$$



Step 2: Geometric Brownian Motion

$$E(\mu_A) = E(\mu_G) > \left(\mu - \sigma^2/2\right)t$$



$$S_T = S_0 e^{rT} \quad \text{- assuming all risk was removed}$$

$$S_T = S_0 e^{\underbrace{\left(r - \sigma^2/2\right)T}_{\text{drift}} + \underbrace{\sigma\sqrt{T}}_{\text{stochastic component}}}$$

- since S_0 is lognormal

$$\ln(S) \sim N\left[\underbrace{\left(1\mu - \frac{\sigma^2}{2}\right)}_{\text{mean}}, \sigma^2\right]$$

ie. growth rate
ie. mean of a geometric return

$$\frac{S_T}{S_0} = e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}}$$

$$\ln\left(\frac{S_T}{S_0}\right) = \left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}$$

$$\ln\left(\frac{S_T}{S_0}\right) \sim \left[N\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]$$

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\frac{\sigma\sqrt{t}}{\sigma}}$$

Step #3: Calculate Z-score

$$Z = \frac{x - \mu}{\sigma}$$

$\mu \rightarrow (r - \sigma^2/2)t$ - what S will grow at

$x = \ln(k/s)$ - what S needs to grow at to = K
(50/40)

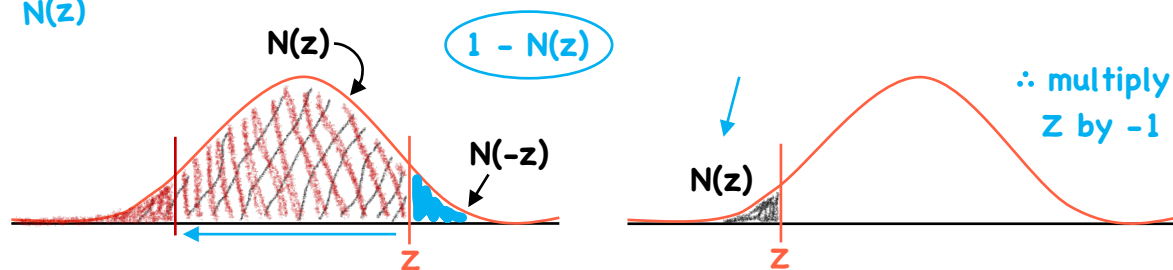
$$Z = \frac{\ln(K/S) - (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$\sigma = \sigma\sqrt{t}$$

if $\text{Var} = \sigma^2 t$

$$\text{sd} = \sigma\sqrt{t}$$

$N(z)$



$$Z = \frac{\ln(K/S) - (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

Note: $\ln(A/B) = \ln(A) - \ln(B)$

$$Z = \frac{\ln(k) - \ln(s) - (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

x - 1

$$Z = \frac{-\ln(k) + \ln(s) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$Z = \frac{\ln(s) - \ln(k) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

Note: $\ln(A) - \ln(B) = \ln(A/B)$

$$d_2 = \frac{\ln(S/K) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

$N(d_2)$ prob. ($S_T \geq K$)

$$d_2 = d_1 - \sigma\sqrt{t}$$

$$d_1 = d_2 + \sigma\sqrt{t}$$

$$d_1 = \frac{\ln(S/K) + (r - \sigma^2/2)t}{\sigma\sqrt{t}} \quad \text{(+ } \sigma\sqrt{t} \text{)} \rightarrow \frac{\sigma^2 t}{\sigma\sqrt{t}} = \frac{\sigma\sqrt{t} \cdot \sigma\sqrt{t}}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln(S/K) + (r - .5\sigma^2)t + \sigma^2 t}{\sigma\sqrt{t}} \quad \frac{1}{2}\sigma^2 \quad .5\sigma^2$$

$$d_1 = \frac{\ln(S/K) + rt - .5\sigma^2 t + \sigma^2 t}{\sigma\sqrt{t}} \quad d_1 = d_2 + \sigma\sqrt{t}$$

$$d_1 = \frac{\ln(S/K) + rt + .5\sigma^2 t}{\sigma\sqrt{t}}$$

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$P(S_T | S_T \geq K)$
payoff $S_T - K$

$\therefore c = S \cdot N(d_1) - Xe^{-rT} N(d_2)$

$\pi = \frac{(1+r) - d}{u - d}$

24

3 x 4 x 5
4 x 3 x 5
5 x 3 x 4

$c = S \cdot N(d_1) - Xe^{-rT} N(d_2)$
why different?

$c = \frac{\pi c^+ + (1 - \pi)c^-}{e^r}$

$c = e^{-r}[\pi c^+ + \underbrace{(1 - \pi)c^-}_0]$

$c = e^{-r}\pi c^+$

$c = e^{-r}\pi(Su - X)$

$c = e^{-r}\pi Su - e^{-r}\pi X$

$c = Se^{-r}\pi u - Xe^{-r}\pi$

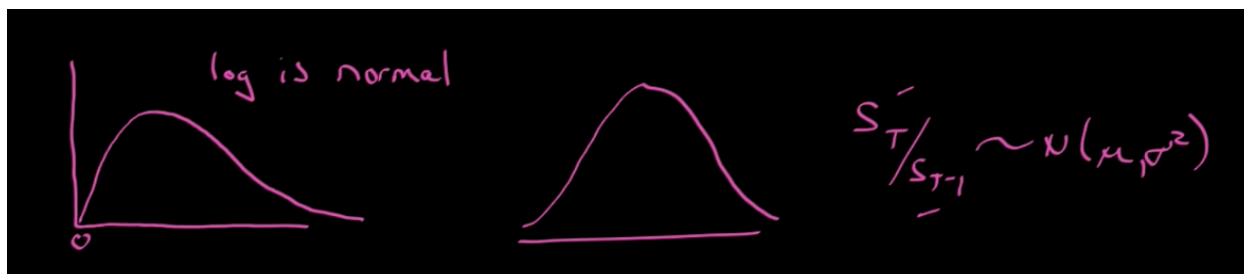
$\pi u > \pi$

$(1+r)^T$

$c^+ = S_T - x = Su - X$

Why do we multiply sd by the square root of T

daily	weekly	monthly	yearly
μ	$\mu(5)$	$\mu(20)$	$\mu(250)$
σ	$\sigma\sqrt{5}$	$\sigma\sqrt{20}$	$\sigma\sqrt{250}$



$S_T = S_0 e^{rT}$

$S_T / S_0 = e^{rT}$

$\ln(S_T / S_0) = \ln(e^{rT})$

$\ln(S_T / S_0) = rT$

$S_T / S_0 = (S_T / S_{T-1}) (S_{T-1} / S_{T-2}) \dots (S_1 / S_0)$

$24 = 6 \times 4$

$\ln(24) = \ln(6) + \ln(4)$

$\ln(S_T / S_0) = \ln(S_T / S_{T-1}) + \ln(S_{T-1} / S_{T-2}) + \dots + \ln(S_1 / S_0)$

$rT = r_{T-1} + r_{T-2} + r_{T-3} + \dots + r_1$

$\downarrow$

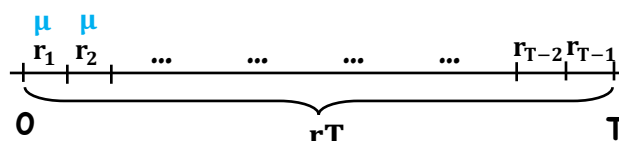
$\sim N(\mu, \sigma^2)$

$E(rT) = \mu + \mu + \mu + \dots + \mu$

$E(rT) = \mu T$

$$rT = r_{T-1} + r_{T-2} + r_{T-3} + \dots + r_1$$

$\sim N(\mu, \sigma^2)$



$$\begin{aligned} \text{Var}(rT) &= \text{Var}(r_{T-1}) + \text{Var}(r_{T-2}) + \text{Var}(r_{T-3}) \dots \text{Var}(r_1) \\ &= \sigma^2 + \sigma^2 + \sigma^2 \dots \sigma^2 \end{aligned}$$

$$\text{Var}(rT) = \sigma^2 T$$

$$\sigma(rT) = \sigma\sqrt{T}$$

$$\sigma^2 = 400 \quad T = 10$$

$$\text{Var}(rT) = \sigma^2 T = 4000$$

$$\sigma(rT) = \sqrt{4000} = 63.2455$$

$$\sigma(rT) = \sigma\sqrt{T} = 20\sqrt{10} = 63.2455$$

$$\ln\left(\frac{S_T}{S_{T-1}}\right)$$

Volatility

sd of the CC returns

- compute σ_{daily}
- weekly $\sigma_{\text{daily}}\sqrt{5}$
- monthly $\sigma_{\text{daily}}\sqrt{20}$

$$\text{VaR} = (E(R_p) - Z \cdot \sigma_p)(-1)NA$$

μ

$$(E(R_p) \cdot 250 - Z \cdot \sigma_p \sqrt{250})(-1)NA$$

μT

$(\sigma\sqrt{T})$

$\sigma^2 T$

VaR

12 LOS explain - 3 compare - 1 estimate - 1
 describe - 6 demonstrate - 1 interpret - 1

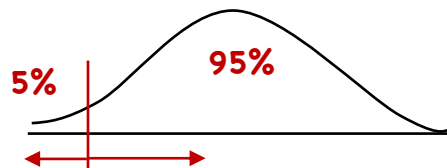
- Meant to capture market risk
 - equity prices
 - interest rates
 - exchange rates
 - commodity prices
- a single security can be a risk factor itself.

e.g./ ① 95% of the time, max. loss will be less than \$1

or ② 5% of the time, min. loss will be at least \$1

- not expected value

5% VaR = \$1M
over a one-day
period



- given assumed market conditions

3 methods to estimate VaR - parametric (var. - covar.)
- historical
- Monte Carlo

Step #1 is always risk decomposition:

- simple → complex - global equities
- each stock on its own - complex bonds (global)
- derivatives

Step #2 - data history - return history

regulation: at least ~ 2 yrs. daily returns
1 yr. daily, updated (updated weekly)
weekly - lookback period

avg. return - daily annual = daily $\times$ 250
s.d. - daily = daily $\times \sqrt{250}$

Parametric/ - assumes return distributions are normal

$$\left(\begin{aligned} E(R_p) &= w_1 E(R_1) + w_2 E(R_2) + \dots \\ \sigma_p &= \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \dots + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2 + 2w_1 w_3 \rho_{13} \sigma_1 \sigma_3 \dots} \end{aligned} \right)$$

need: n $E(R_i)$ $\begin{pmatrix} \sigma_1^2 \\ \sigma_2^2 \\ \vdots \\ \sigma_n^2 \end{pmatrix}$
 n σ_1^2
 $n(n-1)$ cov. (i, j)

$$\left(\begin{array}{ll} \text{min.} & +/- \\ \text{5\% VaR} & [(E(R_p) - 1.65\sigma_p)(-1)]NA \\ \text{1\% VaR} & [(E(R_p) - 2.33\sigma_p)(-1)]NA \end{array} \right) \begin{array}{l} 95 \\ 99 \end{array}$$

1. The parameters of normal distribution required to estimate parametric VaR are:

- A. expected value and standard deviation.
- B. skewness and kurtosis.
- C. standard deviation and skewness.

2. Assuming a daily expected return of 0.0384% and daily standard deviation of 0.9960% (as in the example in the text), which of the following is *closest* to the 1% VaR for a \$150 million portfolio? Express your answer in dollars.

- A. \$3.4 million $[(E(R_p) - z\sigma)(-1)]NA$
- B. \$2.4 million $[(.000384 - 2.33(.00996))(-1)]150M$
- C. \$1.4 million $= 3,423,420$

3. Assuming a daily expected return of 0.0384% and daily standard deviation of 0.9960% (as in the example in the text), the daily 5% parametric VaR is \$2,407,530. Rounding the VaR to \$2.4 million, which of the following values is *closest* to the annual 5% parametric VaR? Express your answer in dollars.

- A. \$38 million
- B. \$35 million
- C. \$600 million

$$\begin{aligned} E(R_p) \times 250 &= .000384 \times 250 = 0.096 \\ \sigma_p \times \sqrt{250} &= .00996 \times \sqrt{250} = 0.15748 \\ [(.096 - 1.65(0.15748))(-1)]150M &= 24,576,653 \end{aligned}$$

summation/

+ simple to understand, normal dist. is easy to work with

-/ very data dependent • reliable?

• VaR is very sensitive to the parameter estimates (μ, σ) • adjustments? • representative?

-/ non-normality is an issue e.g./ options

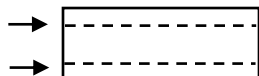
MBS

complex bonds

Historical/

① Reprice current portfolio using historical price series

⇒ output = series of daily returns



rank — high
Low

5% cutoff.

ABC
7 yrs.

Extensions of VaR

1/ Conditional VaR (cVaR)

- the avg. loss 'if' VaR cutoff is exceeded
(expected tail loss, expected shortfall)
- best derived w/ historical simulation
Monte Carlo

2/ Incremental VaR (IVaR)

- how VaR would change if a position size changed (before vs. after)

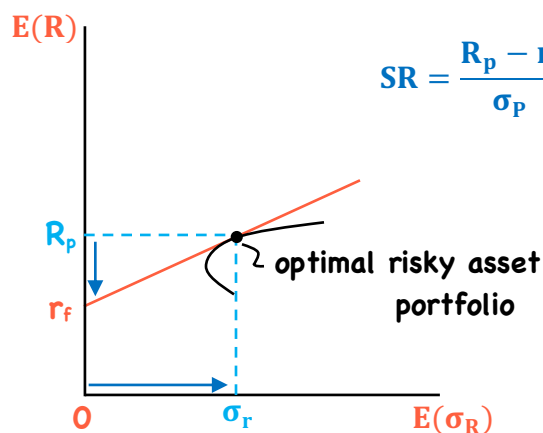
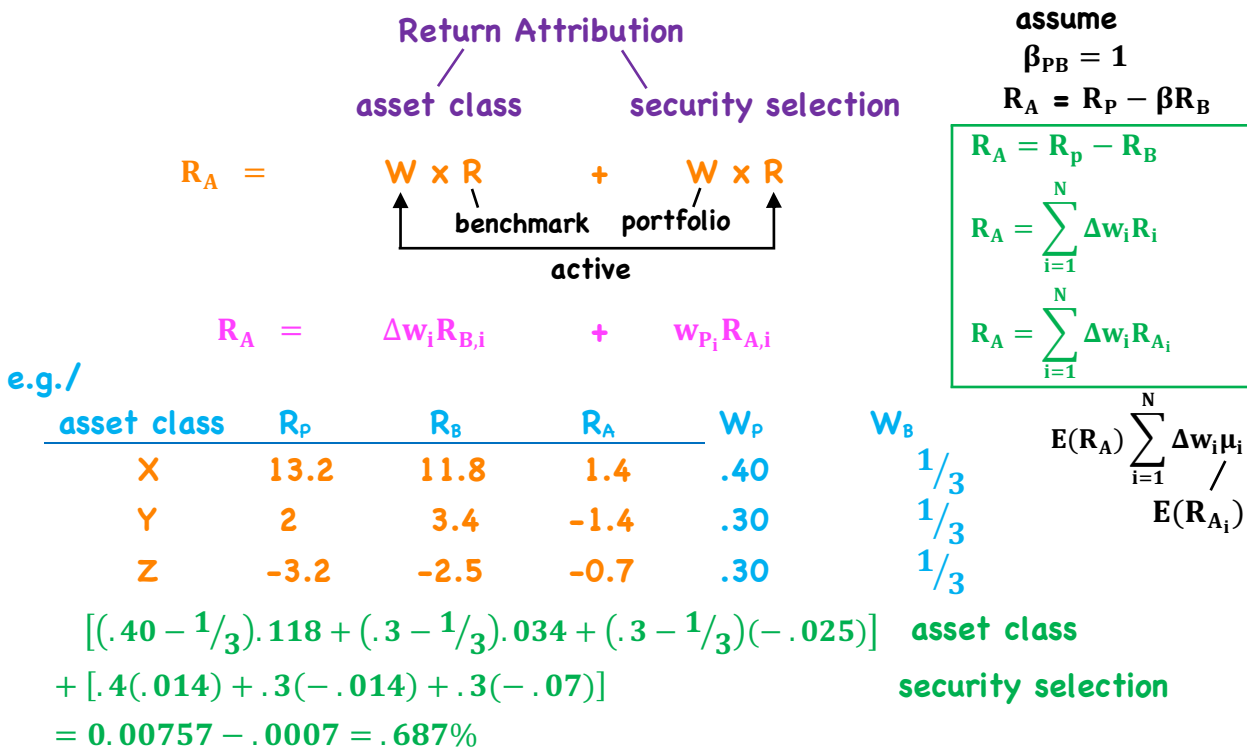
3/ Marginal VaR (MVaR)

- ~ change in VaR for a \$1 or 1%
change in a position

4/ Relative VaR (portfolio positions - benchmark positions)

- ex ante tracking error

Analysis of Active Portfolio Management



- if σ_r is not consistent with investor's risk tolerance, combine the risky portfolio with the risk-free asset with either cash or leverage

- SR_P remains unchanged

